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江苏凯达重工股份有限公司
JIANGSU KAIDA HEAVY INDUSTRY CO. LTD

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型钢轧辊生产基地 出口轧辊生产基地

PROFILE STEEL ROLL PRODUCTION BASE
EXPORT ROLL PRODUCTION BASE

中国·常州·东安
CHANGZHOU DONGAN CHINA



熔炼精品·铸造诚信

SMELTING FINE PRODUCTS AND CASTING INTEGRITY



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COMPANY PROFILE

公司简介

江苏凯达重工股份有限公司，拥有40多年热轧轧辊的研发和制造技术，专业生产冶金轧辊、矿用辊环和高温合金；并为客户提供现场技术服务。公司现有员工430名，其中：研发中心拥有冶炼工艺、金属材料、铸造工艺、热处理工艺、理化检测、加工工艺和轧钢工艺等方面的专业技术人员40多名；公司长期与国内多所院校合作，开展铸造工艺、热处理工艺、加工工艺研究和新产品研发。

江苏凯达重工股份有限公司，公司占地面积9万多平方米，厂房面积7万多平方米。主要生产装备：35KV变电站一座，安装12500KVA计量主变2台，500KVA备用变1台；熔化设备有美国应达公司1-25t感应炉11台套；50t钢包精炼炉1台套；球化处理设备2台套；砂处理设备3台套；铸造除尘系统3台套；热处理炉47台套；烘模炉7台套；卧式离心机10台套；数控和重型加工设备160多台套；75t以下各种规格起重机58台套；检测设备20多台套。年产各类冶金轧辊、矿用辊环和高温合金5万吨，产品最大净重可达50吨。

江苏凯达重工股份有限公司，国家高新技术企业、国家专精特新“小巨人”企业、中国绿色铸造企业、江苏省绿色工厂、江苏省服务型制造示范企业、江苏省企业技术中心、江苏省型钢轧辊工程技术研究中心、江苏省重合同守信用企业、江苏省非公企业党建示范企业、江苏省模范职工之家、江苏省工人先锋号、江苏省优秀“戎耀之家”、江苏省厂务公开民主管理先进单位、江苏省民营科技企业、江苏省名牌产品企业、“凯达牌”商标被认定为江苏省著名商标、江苏省博士后创新实践基地；常州市社会责任标准达标企业、常州市和谐劳动先进企业、常州市安全文明单位、常州市生态园林单位、常州市退役军人就业创业示范基地、常州市星级纳税企业。

江苏凯达重工股份有限公司，中国钢铁工业协会设备分会副会长单位、中国铸造协会轧辊分会轮值理事长单位、全国轧辊企业联盟常务副主席单位；公司已获国家专利局授权专利73件：其中：发明专利30件，实用新型专利43件。公司主持起草了《热轧型钢轧辊》行业标准；参与多项国家标准修订。产品畅销国内外100多家大中型钢铁企业，外销占30%以上；生产规模名列全国同行业前茅，是行业内知名的热轧型钢轧辊和出口轧辊制造企业。

Jiangsu Kaida Heavy Industry Co., Ltd., boasting 40 years of R & D and manufacturing technology of casting hot-rolled rolls, can professionally produce metallurgical rolls, sleeves for mining and high temperature alloy as well as provide on-site technical services to the users. The company has about 430 employees. The R & D center of the company has more than 40 professional technicians specializing in smelting process, metal materials, casting process, heat treatment process, physical and chemical testing, processing process and steel rolling process. The company has long cooperated with many domestic colleges and universities to carry out new product R & D and Research on metal materials and casting technology.

The company covers an area of 90000 m² and a plant area of 70000 m². Main production equipment: one 35kV substation, 2 sets of 12500kVA metering main transformers and 1 set of 500KVA standby transformer. Smelting equipment: 11 sets of 1-25t induction furnaces of Yingda company in the United States; 1 set of 50t ladle refining furnace; 2 sets of spheroidizing treatment equipment; 3 sets of sand treatment equipment; 3 sets of dust removal system; 47 sets of heat treatment furnaces; 7 sets of mould drying furnaces; 10 sets of horizontal centrifuges; More than 160 sets of CNC and heavy processing equipment; 58 sets of cranes of various specifications below 75t; More than 20 sets of testing equipment.

Company is having a production volume of 50000tons of various kinds of rolls and sleeves. The unit finished weight can reach 50tons.

Jiangsu Kaida Heavy Industry Co., Ltd. is a national high-tech enterprise, a national specialized and special new "little giant" enterprise, a Chinese green casting enterprise, Jiangsu Green Factory, a serviceoriented manufacturing demonstration enterprise in Jiangsu Province, Jiangsu enterprise technology center, Jiangsu section steel roll Engineering Technology Research Center, a contract abiding and trustworthy enterprise in Jiangsu Province, Jiangsu non-public enterprise party building demonstration enterprise, Jiangsu model employee's home, Jiangsu private science and technology enterprise, Jiangsu model employee's home, Jiangsu Worker Pioneer, Jiangsu Supreme "Honoring army family", Jiangsu Advanced unit of open and democratic management of factory Affairs, Jiangsu private science and technology enterprise, Jiangsu famous brand product enterprise, "Kaida brand" trademark is recognized as a famous trademark in Jiangsu Province; Jiangsu postdoctoral innovation practice base, Changzhou enterprises meeting social responsibility standards, Changzhou advanced enterprise of harmonious labor, Changzhou safe and civilized unit and Changzhou ecological garden unit, Changzhou veterans employment and entrepreneurship demonstration base, Changzhou Star Level Tax Paying enterprise.

Jiangsu Kaida Heavy Industry Co., Ltd. is the vice chairman unit of equipment branch of China Iron and Steel Industry Association, the rotating chairman unit of roll branch of China Foundry Association and the executive vice chairman unit of national roll enterprise alliance; The enterprise has obtained 73 patents authorized by the State Patent Office, including 30 invention patents and 43 utility model patents. The company presided over the drafting of the industrial standard of hot rolled section steel roll and participated in revisions to various Nation Standard. The products sell well in more than 100 large and medium-sized iron and steel enterprises at home and abroad, and the export of products accounts for more than 30%. The production scale ranks among the top in the same industry in China. It is a well-known hot-rolled section steel roll manufacturing enterprise in the industry.

ORATION OF THE PRESIDENT

董事长致辞



董事长: 许燕楠 Chairman: Xu Yanan

江苏凯达重工股份有限公司的历史，最早可追溯到1952年武进县东安乡政府组建的东安铁木业社；1979年开始生产热轧轧辊；1998年底改制为民营企业；2007年在工业集中区成立合资公司；2021年改制为股份有限公司。70年来努力奋斗，40多年精益求精，经过几代人的不懈努力，企业在行业中确立了地位，在用户中树立了品牌。我们的每一次变革，每一点进步，都离不开合作伙伴和社会各界的关心和支持，更离不开全体员工的辛勤工作，借此机会，我表示衷心的感谢！并真诚地希望你们一如既往关心和支持我们，更期望你们到我们企业莅临指导，我们将不胜荣幸。

展望未来，我们的发展充满挑战，但我们坚信，只要我们为用户提供最适用的产品，只要我们坚持“熔炼精品，铸造诚信”的宗旨，我们就一定能在用户中得到信任，在行业中得到认可，我们将致力于成为百年企业，为中国热轧型钢轧辊行业发展，作出应有的贡献。

让我们携手共进，创造更加辉煌的未来！

The history of Jiangsu Kaida Heavy Industry Co., Ltd. can be traced back to Dong'an iron wood industry society established in 1951; In 1979, it began to produce hotrolled rolls; At the end of 1998, the enterprise was transformed to be a private enterprise; In 2007, a joint venture was founded in industrial concentration area; in 2021, company was restructured to be a company limited by shares. With 70 years of excellence and thanks to the unremitting efforts of several generations, the enterprise has established its position in the industry and its brand among users. Every change and progress we have made can neither be separated from the care and support of all sectors of society and partners, nor from the hard work of all employees. I take this opportunity to express my heartfelt thanks! And I sincerely hope that you will always care about and support us. We will be extremely honored to have your visit and guidance to our enterprise.

Looking forward to the future, our development is full of challenges. But we firmly believe that as long as we provide users with the most applicable products, we will continue to earn trust and confidence from our clients and maintain our position in the industry. As long as we adhere to the purpose of "smelting high-quality products and casting integrity", we will be committed to grow to be a century-old enterprise and make contributions to the development of China's profile steel roll industry.

Let's work together to create a more brilliant future!

凯达重工

凝聚历史积淀底蕴，肩负行业发展重任
不断研发新品替代进口
为钢铁工业发展贡献力量

KAIDA HEAVY INDUSTRY
CONDENSE THE HISTORICAL ACCUMULATION AND SHOULDER THE
IMPORTANT TASK OF INDUSTRY DEVELOPMENT
CONTINUOUSLY DEVELOP NEW PRODUCTS TO REPLACE IMPORTS
CONTRIBUTE TO THE DEVELOPMENT OF IRON AND STEEL INDUSTRY





**WE ARE
ALL KAIDA**
我们都是凯达人



**砥砺前行，聚沙成塔
牢记担当责任使命，携手共创美好世界**

不负韶华，同心同向，沿着热轧型钢轧辊更专、更精的发展目标，砥砺前行。
共建共享，主动担责，回馈社会，造福员工，为一切更美好的梦想延续而奋斗。

**FORGE AHEAD AND GATHER SAND INTO A TOWER
REMEMBER TO TAKE RESPONSIBILITY AND MISSION
JOIN HANDS TO CREATE A BETTER WORLD**

Live up to youth, work together in the same direction, and forge ahead along the development goal of more specialized, refined and special profile steel rolls.
Build and share, take the initiative to shoulder responsibilities, give back to the society, benefit employees, and strive for the continuation of all better dreams.



发展历程 DEVELOPMENT HISTORY

- 1952** 1952年 武进县东安乡政府组建东安铁木业社，生产铁木农具和木制家具产品。
In 1952, the Dong'an Township Government of Wujin County established the Dong'an Iron Wood Industry Association to produce iron wood agricultural tools and wooden furniture products.
- 1971** 1971年 东安铁木业社更名为东安农机具修造厂，开发生产农机配件和机械产品。
In 1971, Dong'an Iron Wood Industry Society changed its name to Dong'an Agricultural Machinery Repair Factory, which developed and produced agricultural machinery parts and machinery products.
- 1979** 1979年12月 武进县东安公社新办武进县东安轧辊厂，开发生产铸铁轧辊系列产品。
In December 1979, the Dong'an Commune of Wujin County set up a new Wujin County Dong'an Roll Factory to develop and produce cast iron roll series products.
- 1983** 1983年05月 东安轧辊厂实行县社联办，更名为武进轧辊厂。
In May 1983, Dong'an roll factory implemented the county social joint office and changed its name to Wujin roll factory.
- 1994** 1994年03月 同意增挂武进县凯达铸钢轧辊厂，开发生产铸钢轧辊产品。
In March 1994, it was agreed to add Wujin Kaida Cast Steel Roll Factory to develop and produce cast steel roll products.
- 1998** 1998年12月 武进轧辊厂、武进市凯达铸钢轧辊厂改制为民营企业。
In December 1998, Wujin Roll Factory and Wujin Kaida Cast Steel Roll Factory were transformed into private enterprises.
- 2001** 2001年05月 武进市凯达铸钢轧辊厂兼并武进轧辊厂。
In May 2001, Wujin Kaida Cast Steel Roll Factory merged with Wujin Roll Factory.
2001年12月 武进市凯达铸钢轧辊厂更名为常州市凯达轧辊有限公司，后更名为常州凯达轧辊集团有限公司。
In December 2001, Wujin Kaida cast steel roll factory was renamed Changzhou Kaida roll Co., Ltd. Later renamed Changzhou Kaida Roll Group Co., Ltd.
- 2006** 2006年05月 常州凯达轧辊集团有限公司投资建设新厂区（东安兴旺路66号），开发生产热轧型钢轧辊系列产品。
In May 2006, Changzhou Kaida Roll Group Co., Ltd. invested in the construction of a new plant area (No. 66, Xingwang Road, Dong'an) to develop and produce a series of hot rolled steel roll products.
- 2007** 2007年02月 新厂区成立合资企业常州英凯轧辊制造有限公司，后更名为常州凯达重工科技有限公司。
In February 2007, Changzhou Kaida roll Co., Ltd. was a joint venture with British Roll Manufacturing Co., Ltd. Establishment: Changzhou yingkai Roll Manufacturing Co., Ltd., later renamed Changzhou Kaida Heavy Industry Technology Co., Ltd.
- 2017** 2017年04月 常州凯达轧辊集团有限公司受让合资公司全部股份。
In April 2017, British Roll Manufacturing Co., Ltd. transferred all shares of the joint venture to Changzhou Kaida Roll Group Co., Ltd.
- 2020** 2020年11月 常州凯达轧辊集团有限公司更名为江苏国冶控股有限公司。
In November 2020, Changzhou Kaida Roll Group Co., Ltd. was renamed Jiangsu Guoye Holding Co., Ltd.
- 2021** 2021年12月 常州凯达重工科技有限公司股改，更名为江苏凯达重工股份有限公司。
In December 2021, Changzhou Kaida Heavy Industry Technology Co., Ltd. changed its name to Jiangsu Kaida Heavy Industry Co., Ltd.

OUR STRATEGY

我们的战略

我们的产品已畅销国内外主要钢铁企业，并分别与宝武集团、河钢集团、鞍钢集团、津西钢铁集团、天柱钢铁集团、永洋钢铁、安塞乐米塔尔钢铁集团、印度钢铁管理局、英国钢铁集团、俄罗斯耶弗拉兹钢铁集团等国内外知名的大型钢铁企业建立了长期的战略合作关系，为客户创造最大的价值！

Our products have been sold well to major iron and steel enterprises at home and abroad, and have established long-term strategic cooperation with famous large iron and steel enterprises at home and abroad, such as Baowu group, Hegang group, Anshan Iron and Steel Group, Jinxi iron and Steel Group, Tianzhu iron and Steel Group, Yongyang iron and steel group, ArcelorMittal Group, Steel authority of India, British Steel Group, Russian Evraz iron and Steel Group, etc., so as to create the greatest value for customers!



COOPERATIVE PARTNER

合作伙伴

 宝武集团	 河钢集团 HBIS GROUP	 鞍钢集团 ANSTEEL	 本钢集团	 包钢集团 BAOGANG GROUP	 中天钢铁 ZENITH 中天钢铁	 宏昌重工	 日照钢铁 RIZHAO STEEL	 紫竹钢铁	 正丰钢铁
 山钢集团 SD STEEL	 攀钢钢钒 PSV	 马钢股份 MA STEEL	 闽光钢铁	 首钢长治	 安米集团 ArcelorMittal	 印钢总局 सेल SAIL	 英国钢铁 BRITISH STEEL	 耶弗拉兹 EVRAZ NORTH AMERICA	 盖尔道集团 GERDAU
 津西钢铁 JINXI GROUP	 天柱钢铁 TIAN ZHU GROUP	 永洋钢铁	 安泰钢铁 AT	 安钢集团 AN GANG GROUP	 赛尔萨集团 CELSA GROUP	 现代集团 HYUNDAI STEEL	 浦项集团 POSCO 소리없이 세상을 움직입니다 www.posco.co.kr	 塔塔集团 TATA STEEL	 纽柯钢铁 NUCOR

STRONG BRAND & MANUFACTURING

SPECIALIZING IN R&D AND
MANUFACTURING OF
HOT ROLLED SECTION STEEL ROLLS

强品牌、强制造

专注于热轧型钢轧辊的研发和制造



生产装备配置先进，拥有熔炼、铸造、热处理、
理化检测、机加工到成品终检的全流程优质轧辊生产线

Advanced production equipment configuration, including the whole process of
highquality roll production line from smelting,casting heat treatment ,
physical and chemical testing, machining to final inspection of finished products



CASTING AND HEAT TREATMENT EQUIPMENT

铸造、热处理



• 大型钢辊顶注
Top injection of large steel roll



• 总容量达25t × 2美国应达中频炉
Total capacity up to 25t × 2 Yingda medium frequency furnace



• Ø1600x4000离心机
Ø1600x4000 centrifuge



• 大型轧辊喷淬设备
Large roll spray quenching equipment



• 热处理轧辊出炉
Heat treatment roll discharge



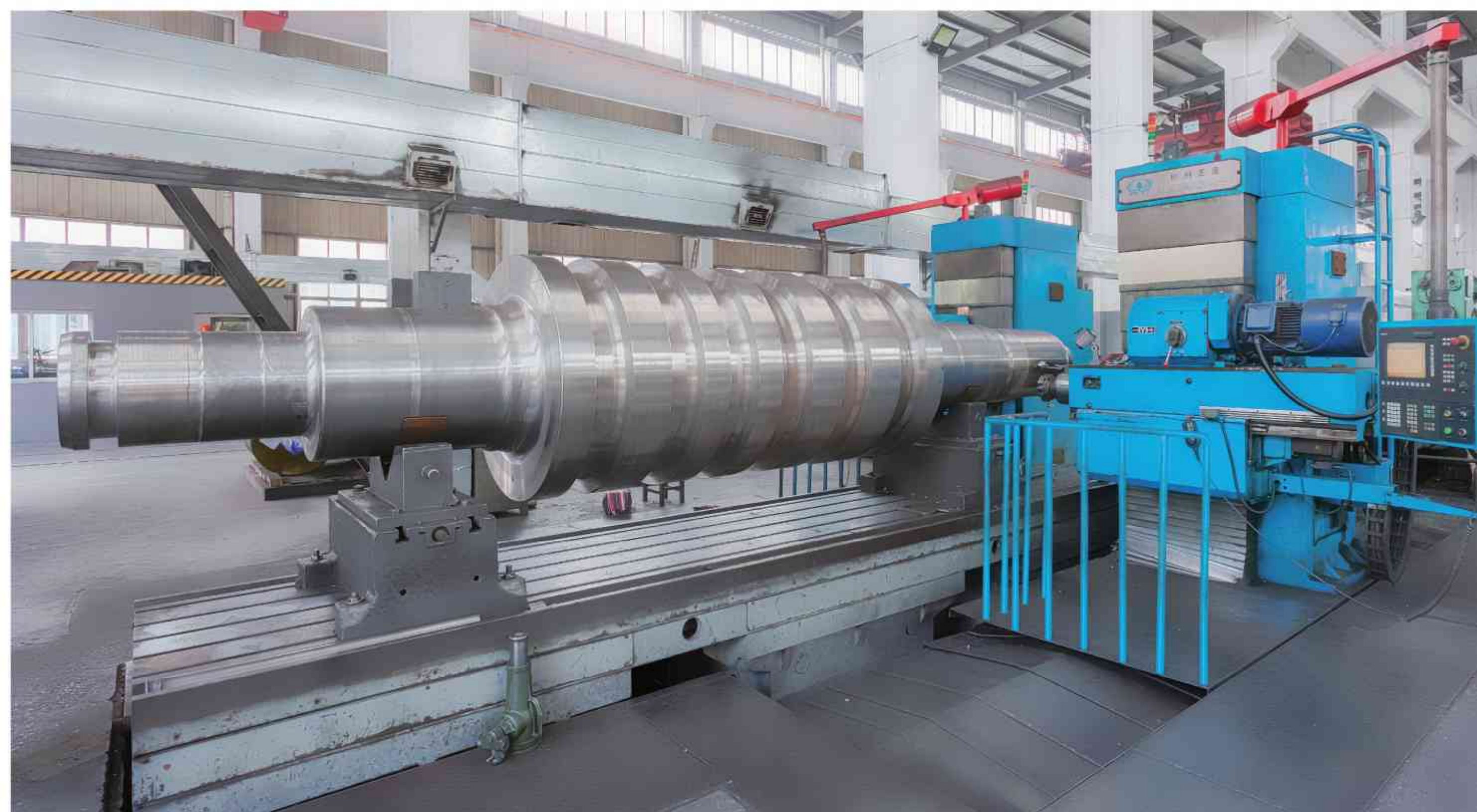
• 拥有50多台热处理设备
It has more than 50 heat treatment equipment



• 精加工车间
Finishing workshop

PROCESSING EQUIPMENT

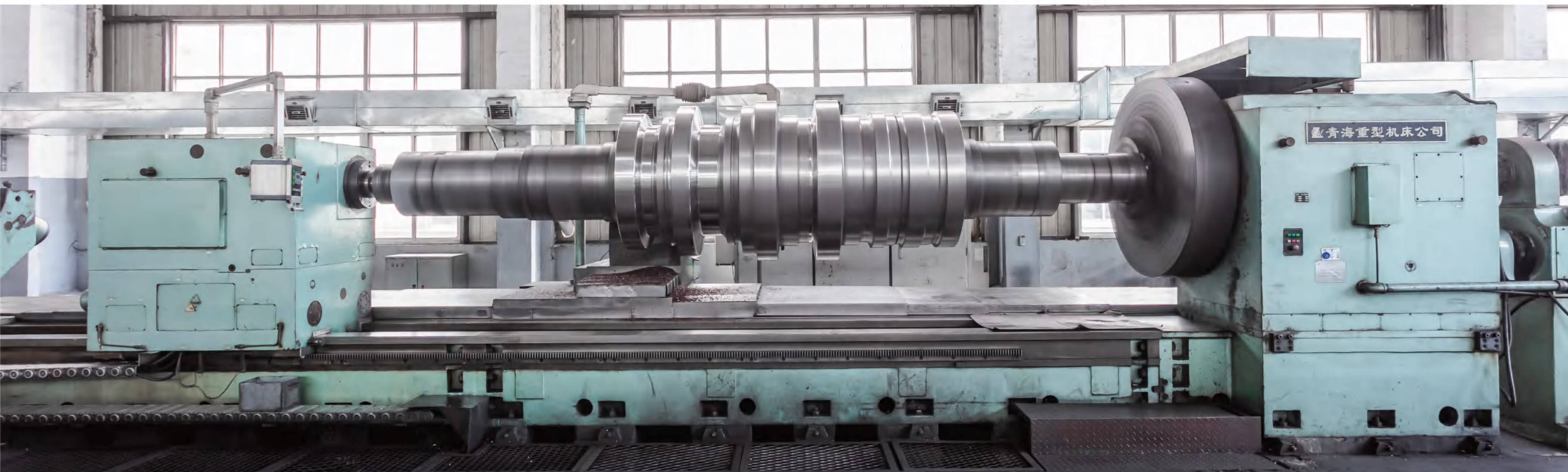
加工设备



• 数控双面铣床
CNC double face milling machine



• 数控镗铣床
CNC boring and milling machine



• 大型数控车床
Large CNC lathe



• 小型数控车床
Small CNC lathe



• 大型数控磨床
Large CNC grinding machine



• 大型数控立车
Large CNC vertical lathe

TESTING EQUIPMENT

检测设备



• MES系统信息化质量控制
INFORMATION QUALITY CONTROL OF MES SYSTEM



• 金相实验分析
Metallographic experimental analysis



• 抗拉抗弯试验
Tensile bending test



• 直读光谱分析
Direct reading spectral analysis



• 冲击试验机
Impact testing machine



• 扫描电镜
Scanning electron microscope



• 超声波检测
Ultrasonic testing



• 化学成份调试
Chemical composition commissioning

SEIKO MANAGEMENT

WELL-KNOWN PRODUCTION BASE OF HOT
ROLLED SECTION STEEL ROLLS AT
HOME AND ABROAD AND PRODUCTION
BASE FOR EXPORTING ROLLS

精工艺、精管理

国内外知名的热轧型钢轧辊制造基地
出口轧辊生产基地

引用数字化（MES系统）、信息化（ERP系统）管理
全流程智能严控，确保轧辊品质的可追溯性、稳定性和可靠性

Reference digital (MES system) and information (ERP system) management
The whole process is intelligently and strictly controlled to ensure the traceability,
stability and reliability of roll quality

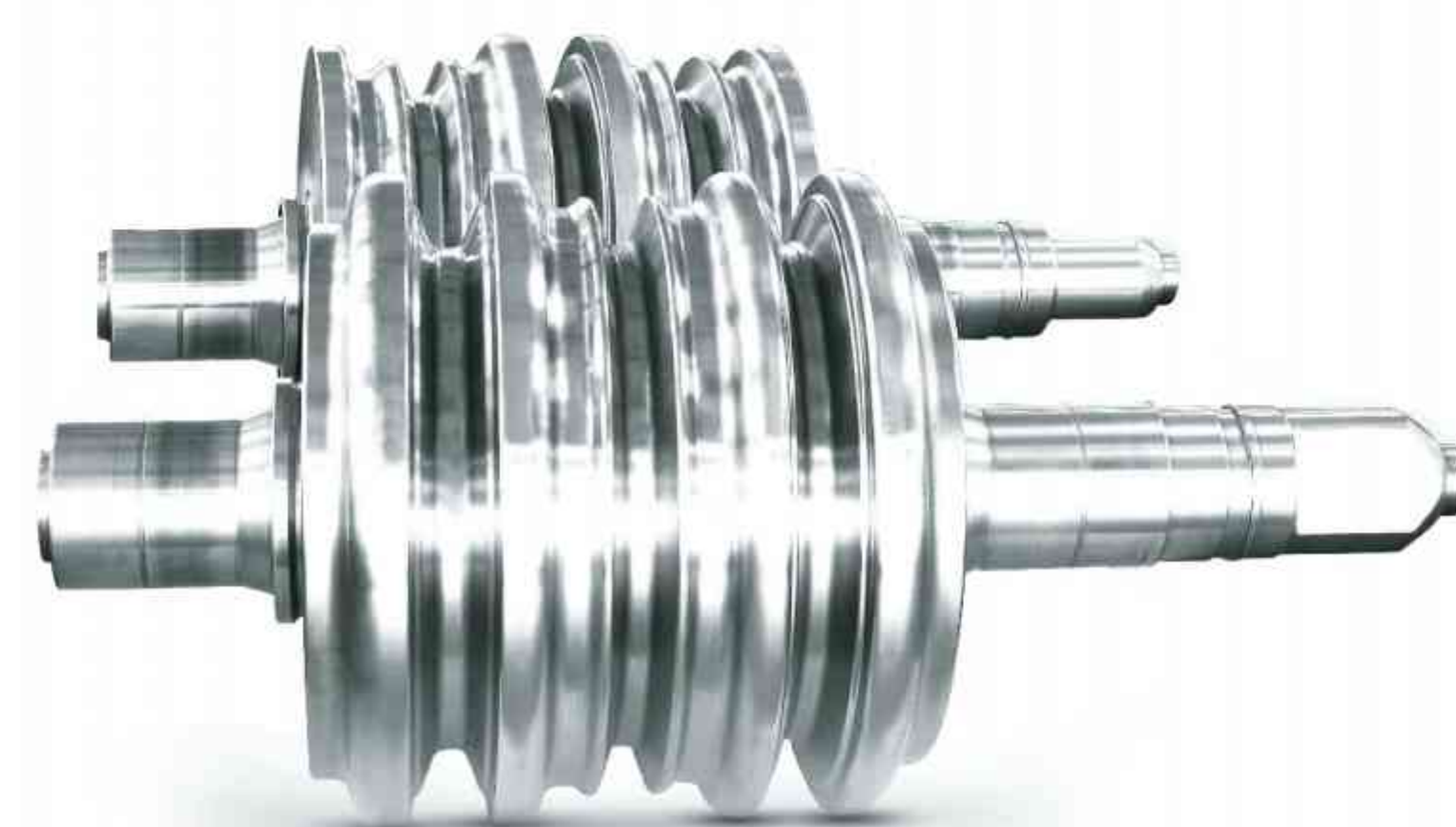


PREMIUM QUALITY PRODUCTS

优质产品



合金铸钢轧辊
Alloy Steel rolls



半钢轧辊
Adamite rolls



石墨钢轧辊
Graphite Steel roll



热作模具钢轧辊
Hot Working Die Steel rolls



高速钢轧辊
High Speed Steel rolls



球墨铸铁轧辊
Spheroidal Graphite Cast Iron Rolls

年产轧辊和辊环5万吨
产品最大净重50吨

ANNUAL OUTPUT OF ROLL AND ROLL
RING IS 50000 TONS
MAXIMUM PRODUCT NET WEIGHT 50 TONNES

PREMIUM QUALITY PRODUCTS

优质产品



NCC轧辊
Non Continuous Carbides Rolls



高强度合金球墨铸铁轧辊
High strength alloy nodular cast iron Rolls



贝氏体球墨铸铁轧辊
Bainitic Nodular Cast Iron Rolls



珠光体球墨铸铁轧辊
Spheroidal Graphite Pearlite Cast Iron Rolls



高铬铸铁轧辊
High Chromium Cast Iron Rolls



合金无限冷硬铸铁轧辊
Alloy Indefinite Chilled Cast Iron Rolls

硬度落差小，抗热裂超耐磨
冲击韧性高，使用寿命长

SMALL HARDNESS DROP, HEAT CRACK
RESISTANCE AND SUPER WEAR
RESISTANCE HIGH IMPACT TOUGHNESS
AND LONG SERVICE LIFE

PREMIUM QUALITY PRODUCTS

优质产品



高铬铸钢轧辊
High Chromium Cast Steel Rolls



锻钢轧辊、辊轴
Forged Steel Rolls Rolls Shaft



轧边辊
Edger Rolls



万能轧机粗、精轧用辊环
Roll ring for rough and finish rolling of universal mill



矿用辊环
Mining Rolls Ring



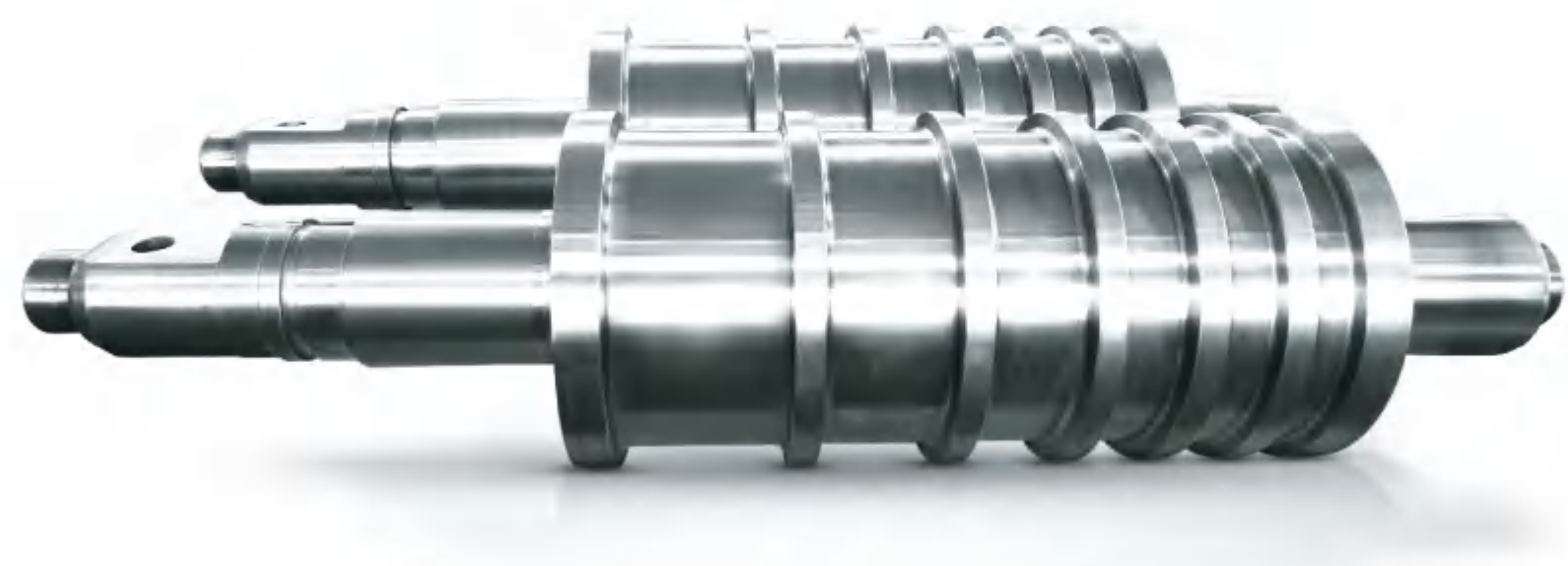
矫直辊
Straightening Rolls

硬度落差小，抗热裂超耐磨
冲击韧性高，使用寿命长

SMALL HARDNESS DROP, HEAT CRACK
RESISTANCE AND SUPER WEAR
RESISTANCE HIGH IMPACT TOUGHNESS
AND LONG SERVICE LIFE

合金铸钢轧辊 / Alloy Steel Rolls

代码(Grade): AS



合金铸钢轧辊是使用中频炉熔炼优质钢水，采用先进的铸造、热处理技术制造，具有很高的抗拉强度、优良的抗热裂性、韧性、耐磨性，适用于型钢粗、中轧机，热轧带钢粗轧辊及支承辊。辊身金相组织为珠光体或回火索氏体。

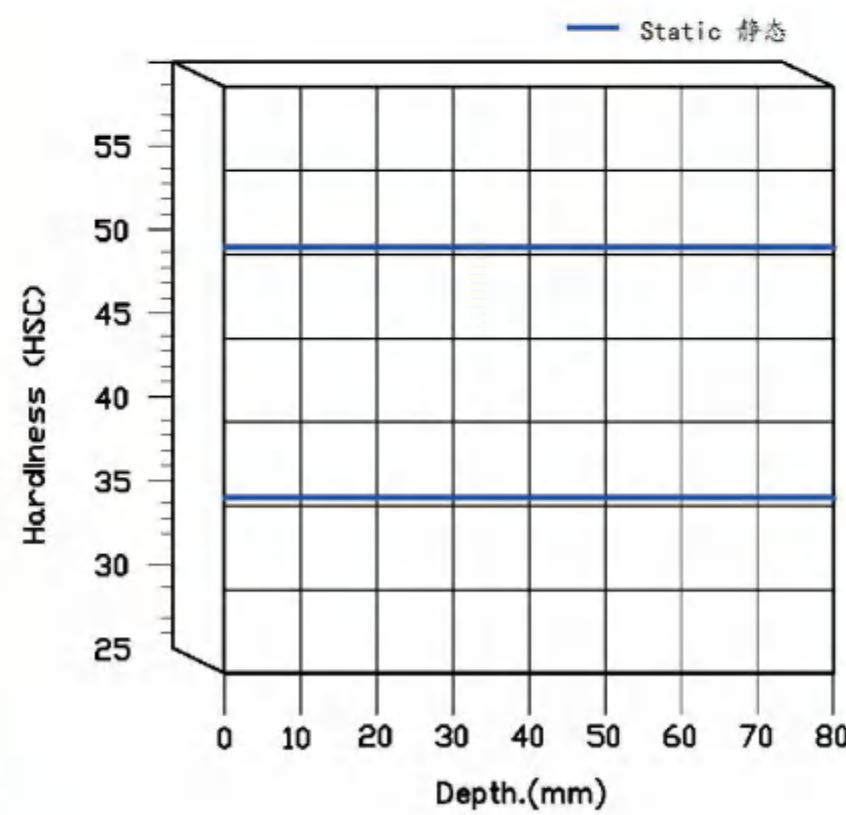
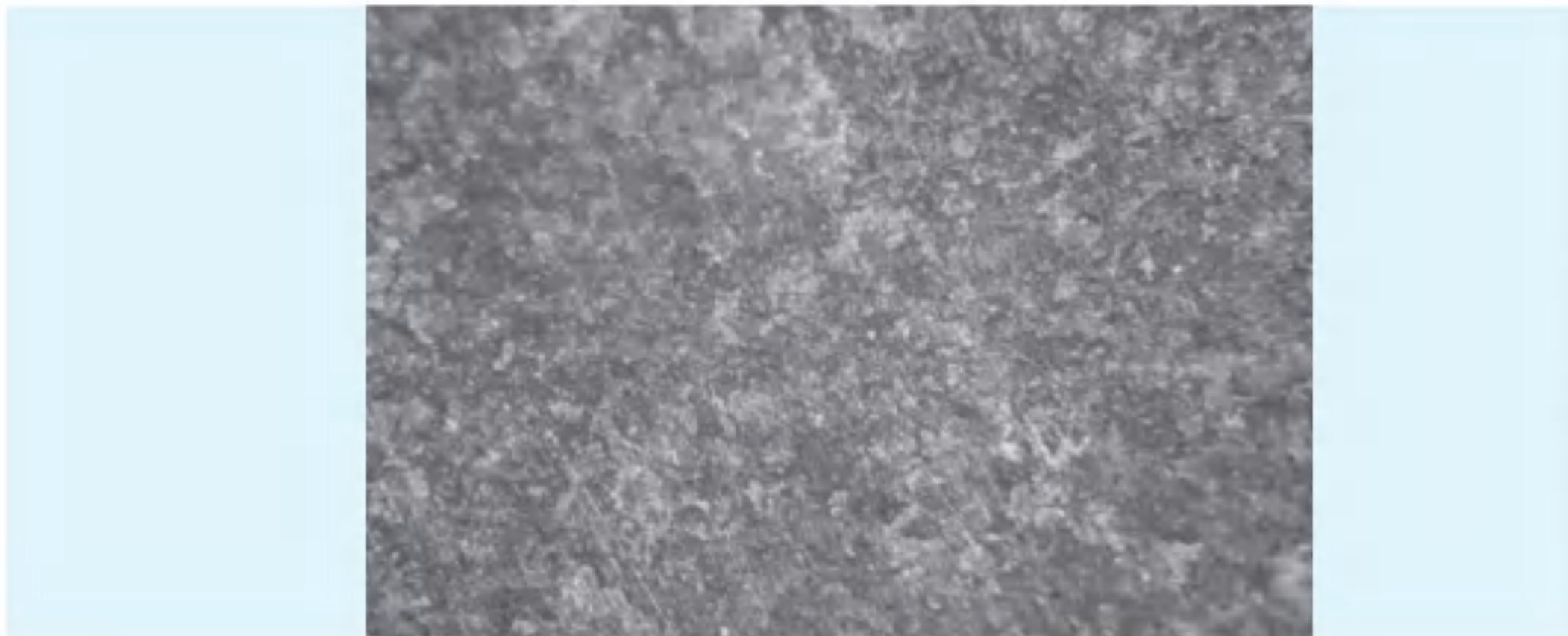
Alloy cast steel roll is made of high quality molten steel melted in medium frequency furnace by advanced casting and heat treatment technology. It has high tensile strength, excellent hot crack resistance, toughness and wear resistance, and is suitable for rough and medium rolling mills of section steel, rough rolls of hot rolled strip steel and backup rolls. The metallographic structure of the roll body is pearlite or tempered sorbite.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)			C	Si	Mn	Cr	Ni	Mo	P	S	V
YB/T 4906	GB/T 1503	自有材质 (Own material)									
AS-650	AS80I/AS90I	—	0.7-1.0	0.2-0.6	0.6-1.0	0.7-3.0	0.3-1.0	0.2-0.6	≤0.03	≤0.03	—
AS-700	AS60-AS75I	—	0.5-0.8	0.2-0.6	0.6-1.0	0.7-1.2	0.3-1.0	0.2-0.6	≤0.03	≤0.03	—
AS-900	AS50I	—	0.4-0.6	0.4-0.7	0.5-1.0	1.5-2.5	1.0-2.0	0.8-1.2	≤0.03	≤0.03	0.2-0.6
AS-1000	AS50	—	0.4-0.6	0.1-0.4	0.5-1.0	0.5-1.5	1.0-2.0	0.2-0.8	≤0.03	≤0.03	—
—	—	Cr3/Cr4/Cr5	0.4-0.7	0.2-0.7	0.7-1.2	2.5-5.5	≥0.5	0.4-1.0	≤0.03	≤0.03	0-0.6

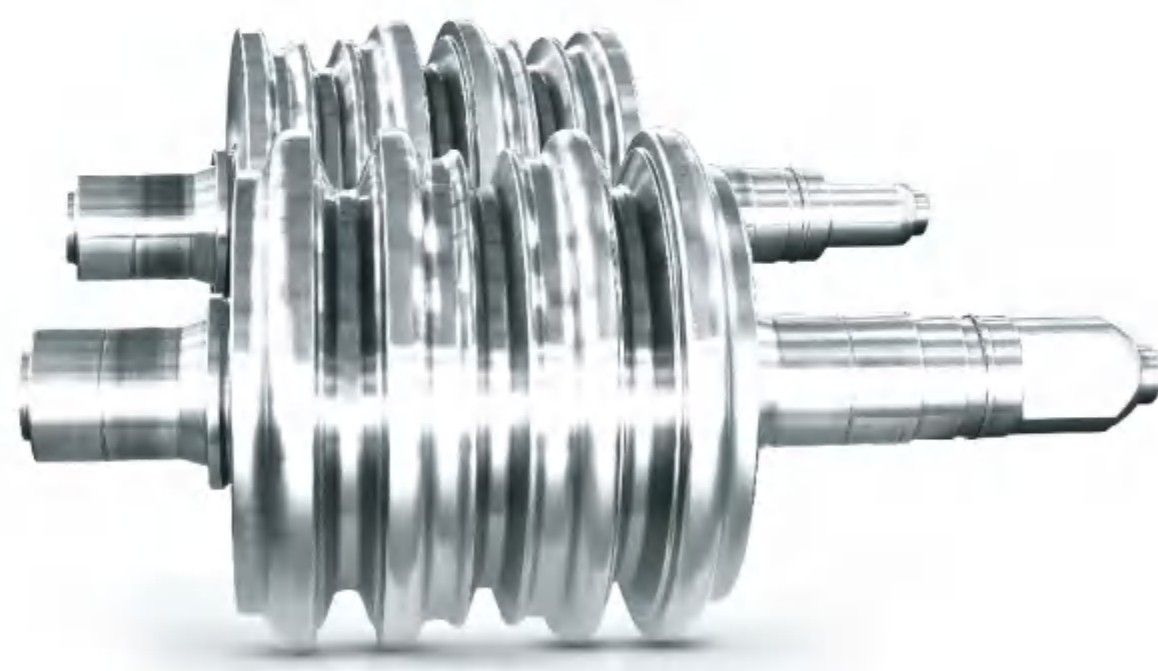
性能及应用 (Properties and Applications)

材质代码 (Material code)			辊身硬度 (HSD)	辊颈硬度 (HSD)	抗拉强度 Rm(N/mm ²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
YB/T 4906	GB/T 1503	自有材质 (Own material)						
AS-650	AS80I/AS90I	—	40-50/45-55	≤45	≥650	φ 1600	50	型钢轧机用粗、中轧辊； 热轧带钢支撑辊、粗轧辊、立辊； Coarse and medium rollers for section mill Hot rolled strip backup roll, roughing roll, vertical roll
AS-700	AS60-AS75I	—	38-48	≤45	≥700			
AS-900	AS50I	—	50-60	≤45	≥900			
AS-1000	AS50	—	40-50	≤45	≥1000			
—	—	Cr3/Cr4/Cr5	55-70	≤45	≥700			



半钢轧辊 / Adamite Rolls

代码(Grade): AD



半钢轧辊机械性能介于铸钢和铸铁之间，根据使用要求采用高温扩散退火、球化退火、正火、回火热处理，其基体组织为索氏体，其中含有块状、颗粒状、细条状的碳化物，有高的耐磨性、强的韧性和好的抗热裂性。最大的特点是在工作层中几乎没有硬度下降。适合热轧带钢连轧机粗轧、精轧前段；型钢轧机粗轧、中轧、预精轧机架，矫直辊；万能轧机辊环。

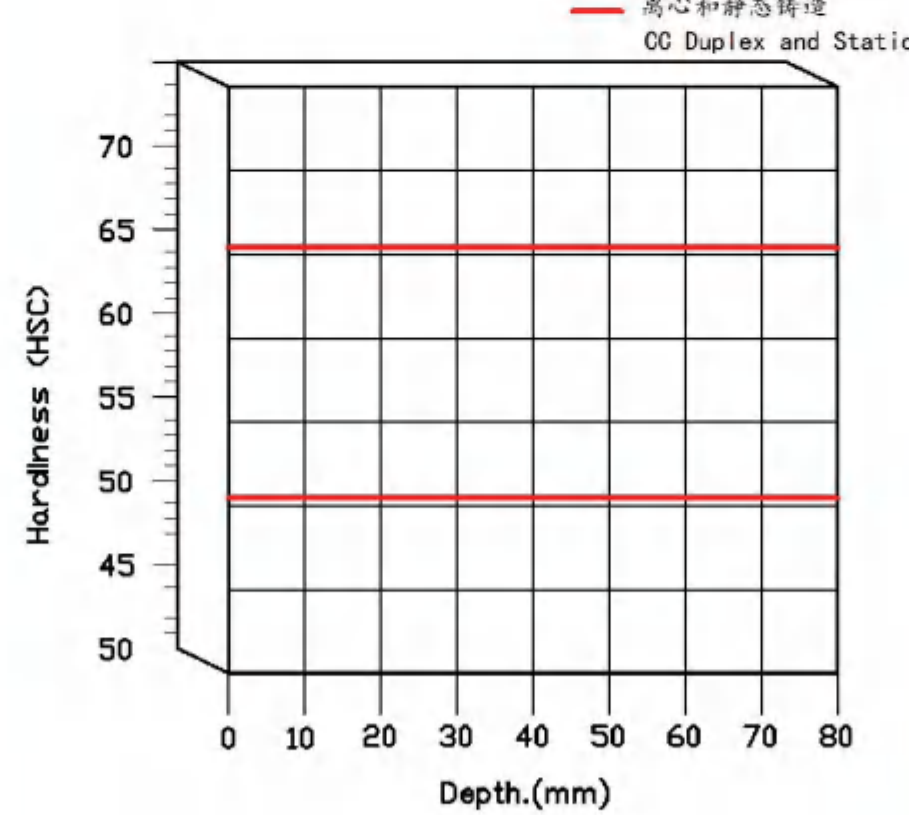
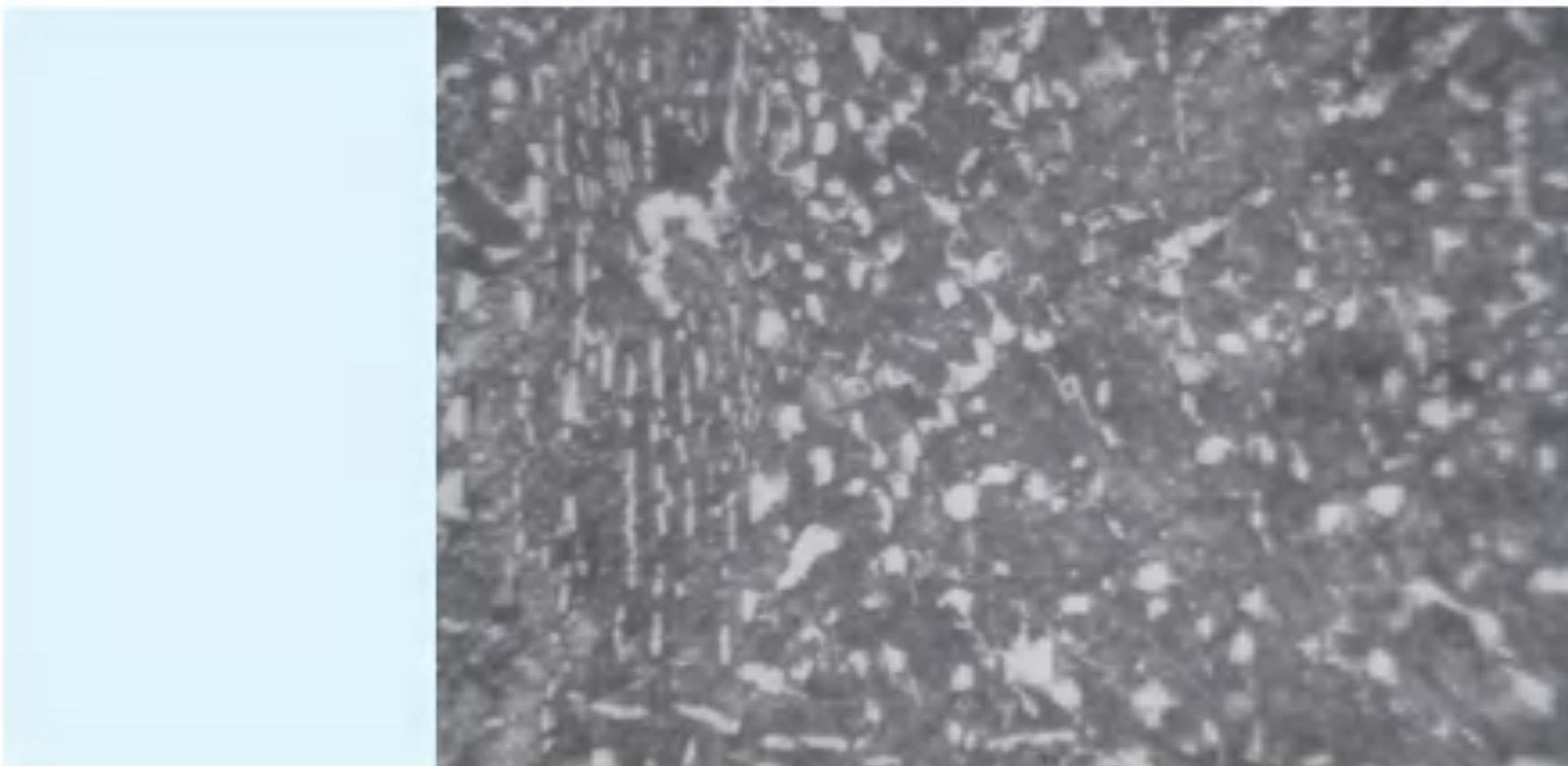
The mechanical properties of semi steel roll are between cast steel and cast iron. High temperature diffusion annealing, spheroidizing annealing, normalizing and tempering heat treatment are adopted according to the use requirements. Its matrix structure is sorbite, which contains massive, granular and fine carbide, and has high wear resistance, strong toughness and good hot crack resistance. The biggest feature is that there is almost no hardness drop in the working layer. Suitable for the front section of roughing and finishing rolling of hot strip mill; Roughing, intermediate and pre finishing stands of section mill, straightening roll; Universal mill roll ring.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)			C	Si	Mn	Cr	Ni	Mo	P	S
YB/T 4906	GB/T 1503	自有材质 (Own material)								
AD-500I	AD160I, AD180-AD200	—	1.5-2.1	0.3-0.8	0.6-1.0	0.7-1.8	0.5-1.8	0.2-0.6	≤0.03	≤0.03
AD-500II	AD180-AD200	—	1.7-2.2	0.3-0.8	0.5-1.5	2.0-4.0	1.0-2.0	0.3-0.6	≤0.03	≤0.03
AD-600	AD140, AD140I	—	1.1-1.5	0.3-0.8	0.6-1.0	0.7-1.7	0.5-1.5	0.2-0.6	≤0.03	≤0.03
—	—	GT	1.4-1.6	0.3-0.8	1.0-2.0	1.0-2.0	1.0-2.0	0.4-1.0	≤0.03	≤0.03

性能及应用 (Properties and Applications)

材质代码 (Material code)			辊身硬度 (HSD)	辊颈硬度 (HSD)	抗拉强度 Rm(N/mm ²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
YB/T 4906	GB/T 1503	自有材质 (Own material)						
AD-500I	AD160I/AD180-AD200	—	40-50/50-60	≤45	≥500	φ 1700	40	型钢轧机用粗、中、精轧辊、矫直辊； 型钢轧机用水平辊环、立辊环； 热轧板带钢粗轧辊、支承辊、立辊； Rough, medium and finishing rolls and straightening rolls for section mills Horizontal and vertical roller rings for section mills Rough roll, support roll and vertical roll of hot rolled strip steel
AD-500II	AD180-AD200	—	55-65	≤45	≥500			
AD-600	AD140I/AD140I	—	40-50/45-55	≤45	≥600			
—	—	GT	50-60	≤45	≥690			



石墨钢轧辊 / Graphite Steel Rolls

代码(Grade): GS



石墨钢轧辊的性能与半钢轧辊类似，其最大特征是组织中存在少量细小球状石墨。它可以提高轧辊的抗热裂性能和抗氧化铁皮粘附性能，对水冷条件要求较低，主要适用于型钢轧机粗轧和初轧机开坯机架。

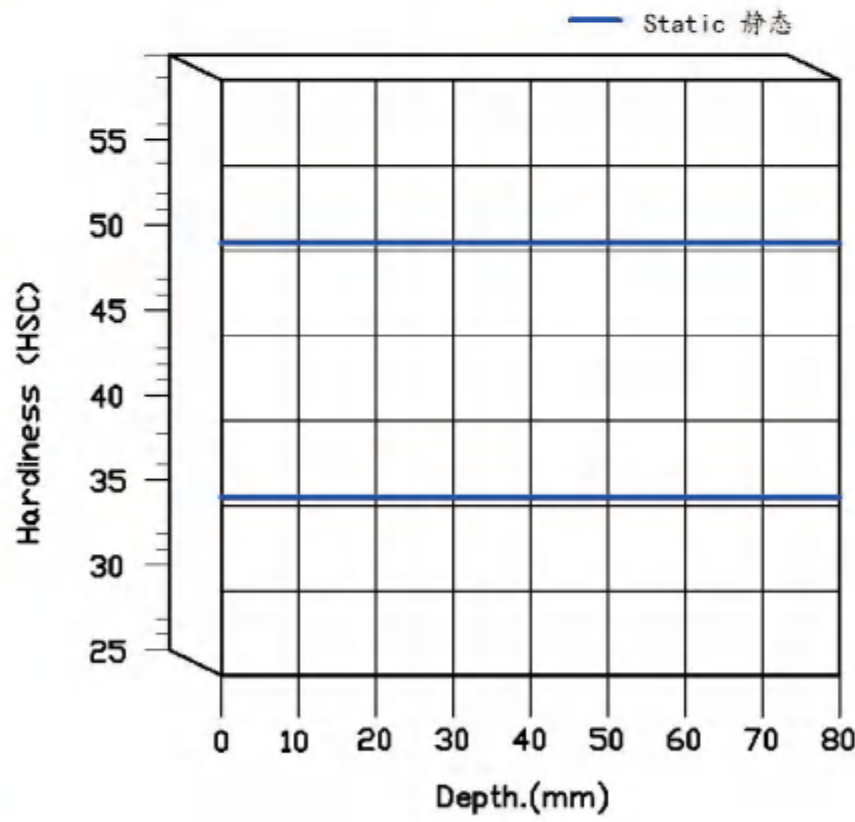
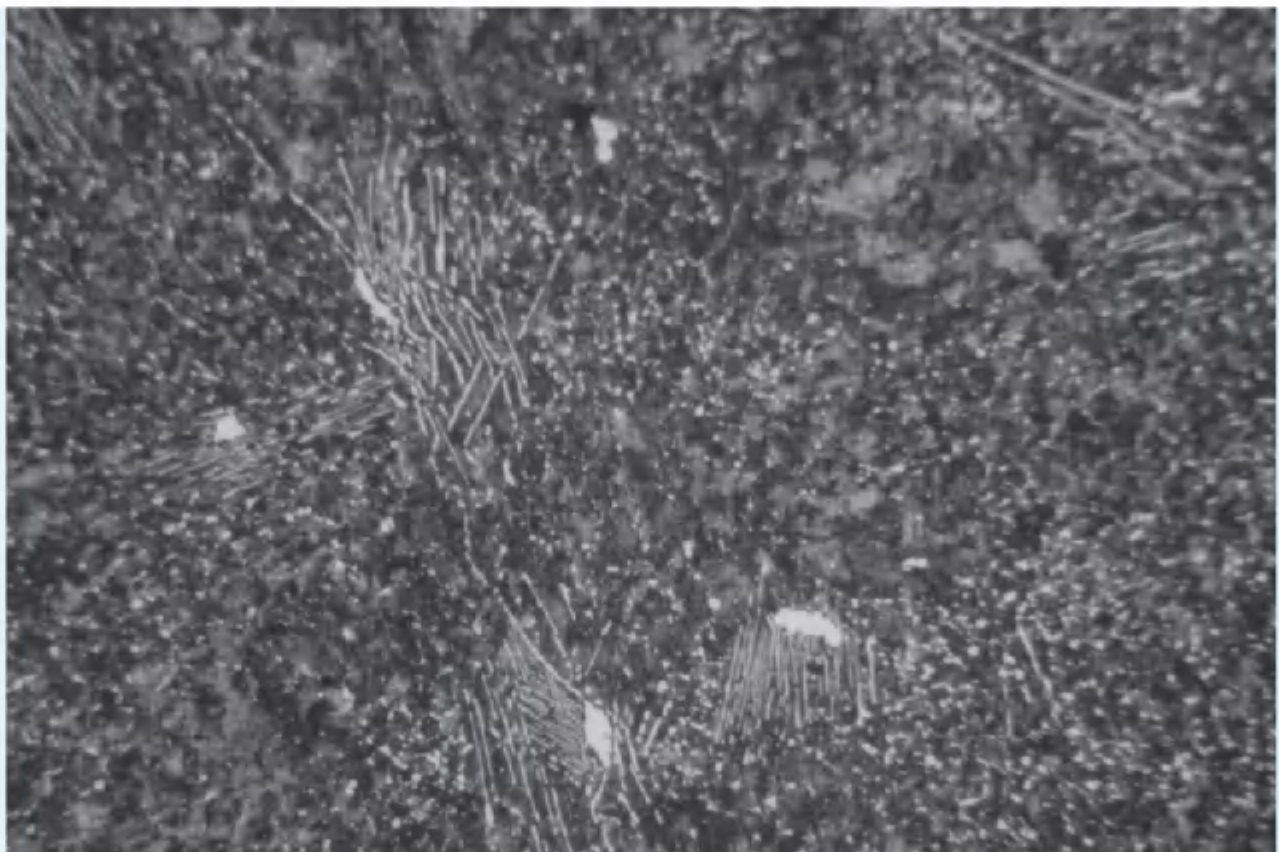
The performance of the graphite steel roller is similar to that of the semi steel roller, and its biggest feature is that there is a small amount of fine spherical graphite in the structure. It can improve the resistance to hot cracking and the adhesion of oxidation resistant iron sheet of the roll, and has low requirements for water cooling conditions. It is mainly applicable to the roughing and blooming stands of the section mill.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)		C	Si	Mn	Cr	Ni	Mo	P	S
YB/T 4906	GB/T 1503								
GS-600	GS140 / GS160	1.2~1.6	1.0~1.8	0.6~1.0	0.7~1.7	0.5~1.0	0.2~0.6	≤0.03	≤0.03
GS-500I	GS160 / GS190 / GS210	1.6~2.1	1.0~1.8	0.6~1.6	1.0~2.0	1.0~2.0	0.4~1.0	≤0.03	≤0.03
GS-500II	GS210 / GS240	2.1~2.5	1.0~1.8	0.5~1.5	1.0~3.0	2.0~3.0	0.5~1.5	≤0.03	≤0.03

性能及应用 (Properties and Applications)

材质代码 (Material code)		辊身硬度 (HSD)	辊颈硬度 (HSD)	抗拉强度 Rm(N/mm²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
YB/T 4906	GB/T 1503						
GS-600	GS140 / GS160	40~50/50~60	≤45	≥600	φ 1300	30	型钢轧机用粗、中、精轧辊、水平辊环立辊环、矫直辊、轧边辊；开坯辊；热轧板带钢粗轧辊、立辊； Rough, medium and finish rolls and horizontal roll rings for section mills Vertical roll ring, straightening roll, edging roll Bloom roll; Rough roll and vertical roll of hot rolled plate and strip steel
GS-500I	GS160 / GS190 / GS210	40~50/50~60	≤45	≥500			
GS-500II	GS210 / GS240	60~70	≤45	≥500			



热作模具钢轧辊具有高热强度、高热稳定性、良好的耐磨性、抗冷热疲劳性能、抗回火稳定性，适用于大型型钢的开坯轧辊、粗轧辊，热轧带钢粗轧辊、支撑辊。

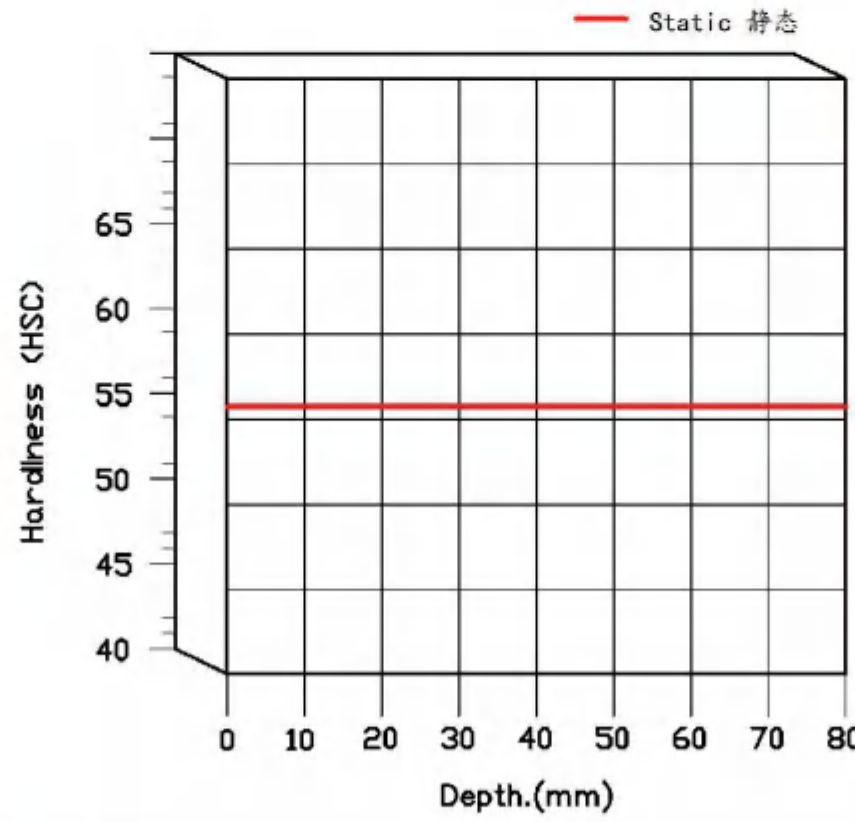
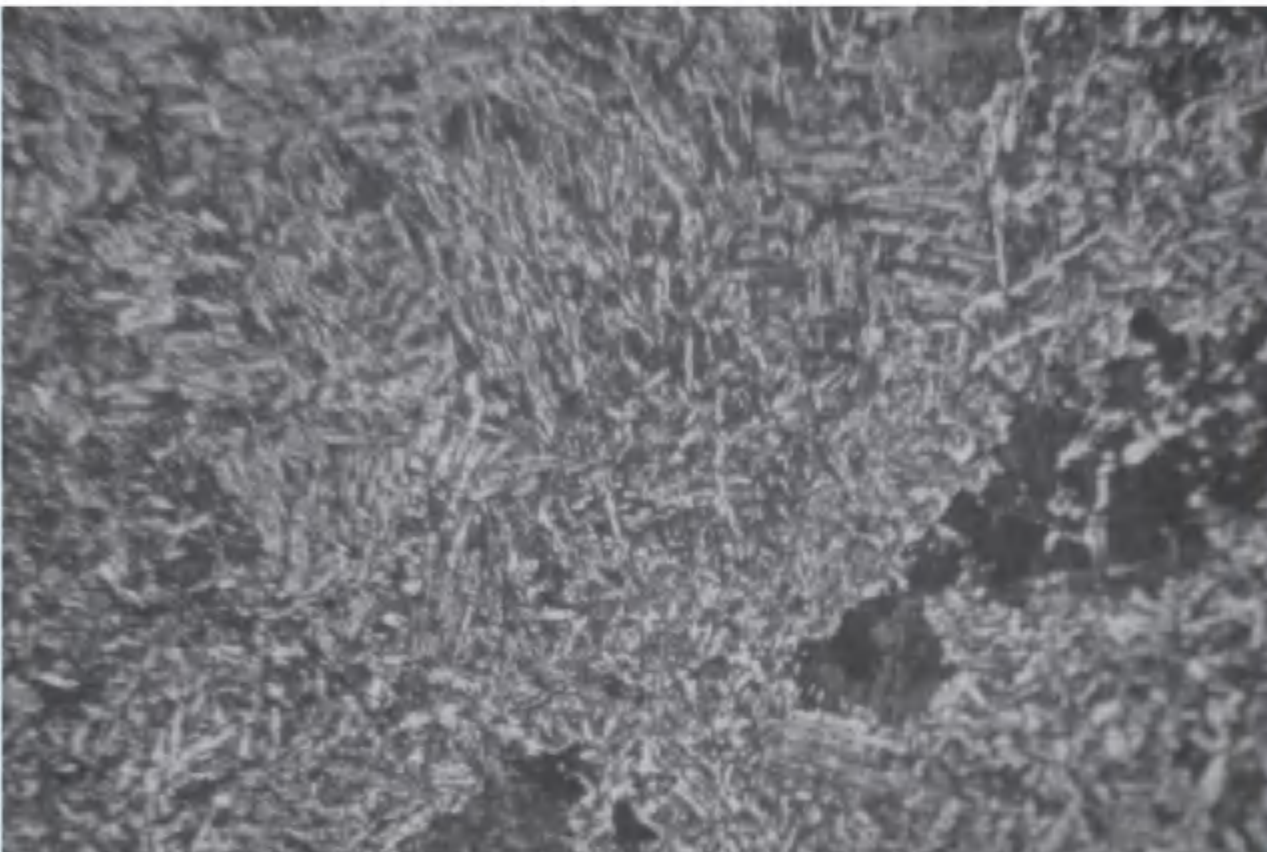
The hot work die steel roll has high thermal strength, high thermal stability, good wear resistance, cold and thermal fatigue resistance, and tempering stability. It is applicable to the cogging roll, rough roll, and backup roll of large section steel.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)	C	Si	Mn	Cr	Ni	Mo	其他
自有材质 (Own material)							
HWD I	0.5~0.6	0.1~0.6	0.3~0.8	0.5~1.0	1.3~1.8	0.1~0.5	—
HWD II	0.4~0.5	0.3~0.8	0.3~0.8	1.5~2.0	0.7~1.2	0.7~1.2	0.3~0.8
HWD III	1.4~1.6	0.1~0.6	0.3~0.8	10.0~14.0	—	0.7~1.2	0.8~1.5

性能及应用 (Properties and Applications)

材质代码 (Material code)	辊身硬度 (HSD)	抗拉强度 Rm(N/mm²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
自有材质 (Own material)					
HWD I	40~50	≥1000	φ 1600	50	大型型钢开坯轧辊、粗轧辊； 热轧带钢粗轧辊、支撑辊； Large section steel bloom roll and roughing roll Rough roll and support roll of hot rolled strip
HWD II	50~60	≥1100			
HWD III	55~65	≥500(core)			万能轧机立辊、轧边辊 Vertical roll and edging roll of universal mill



高速钢轧辊 / High Speed Steel Rolls

代码(Grade): HSS



高速钢轧辊在高温下具有很高的硬度和耐磨性。采用离心方法生产，芯部材质为高强度球墨铸铁。通过成分和热处理工艺控制，工作层硬度可达70-90HSC，马氏体基体上分布有钒、钨、铌、钼复合碳化物，工作层硬度均一，孔型磨损均匀。这种轧辊用于精轧机架，增加轧机作业时间，改善轧材表面质量。

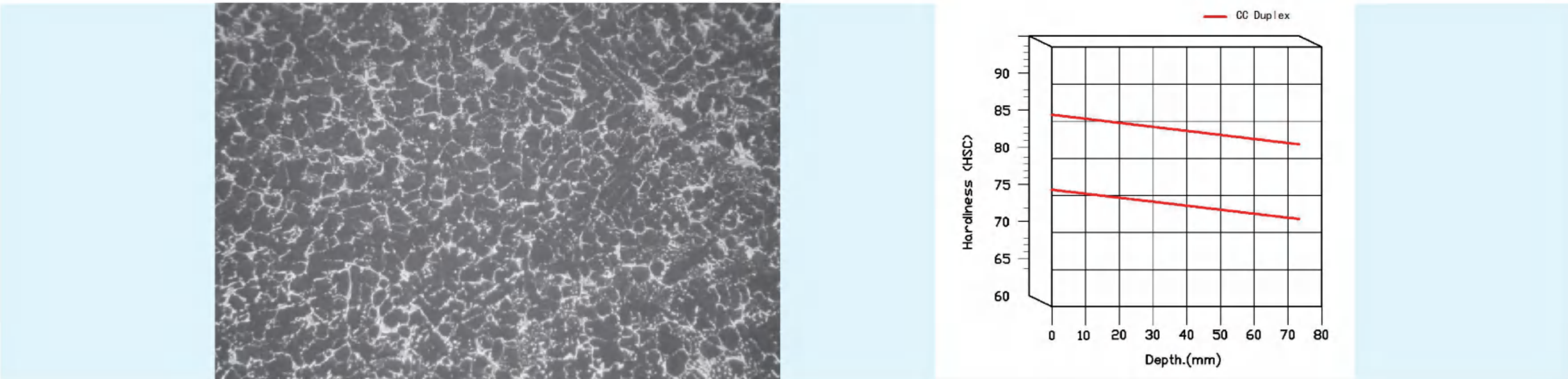
High speed steel roller has high hardness and wear resistance at high temperature. The core is made of high-strength ductile iron by centrifugal method. Through the control of composition and heat treatment process, the hardness of the working layer can reach 70-90HSC, and there are vanadium, tungsten, niobium, molybdenum composite carbides distributed on the martensite matrix. The hardness of the working layer is uniform, and the wear of the groove is uniform. This kind of roll is used in the finishing mill stand to increase the operation time of the mill and improve the surface quality of rolled products.

化学成分(Cheical Compositions) wt. %

材质代码(Material code)			C	Si	Mn	Cr	Ni	Mo	P	S	V	W	Nb	B
YB/T 4906	GB/T 1503	自有材质 (Own material)												
HSS-400	HSS	—	1.5-2.5	0.3-1.5	0.3-1.0	2.0-8.0	0.0-1.5	3.0-8.0	≤0.03	≤0.025	2.0-8.0	0.0-4.0	—	—
—	S-HSS	—	0.5-1.5	0.8-1.5	0.5-1.0	3.0-9.0	0.2-1.2	2.0-5.0	≤0.03	≤0.025	0.4-3.0	0.0-3.0	0.0-2.0	—
—	—	B-HSS	2.0-3.0	0.3-1.5	0.4-1.2	10.0-17.0	1.0-3.0	0.5-2.0	≤0.03	≤0.025	0.1-1.0	—	—	0.1-1.0

性能及应用(Properties and Applications)

材质代码(Material code)			辊身硬度 (HSD)	辊颈硬度 (HSD)	抗拉强度 Rm(N/mm²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
YB/T 4906	GB/T 1503	自有材质 (Own material)						
HSS-400	HSS	—	70-90	35-45	≥400	φ 650	5	中小型型钢轧机用中、精轧辊、水平辊环、立辊环；棒线材粗、中、精轧； Vertical roll ring; Coarse, medium and finish rolling of rods and wires; Hot rolled strip steel precision roll;
—	S-HSS	—	70-85	35-45	≥400			
—	—	B-HSS	75-85	35-45	≥400			



球墨铸铁轧辊 / Spheroidal Graphite Cast Iron Rolls (Nodular Cast Iron)

代码(Grade): SG



球墨铸铁材质的特征是铸态下石墨呈球状，它的性质与合金无限冷硬轧辊相似，其强度高于无限冷硬轧辊。

The feature of ductile iron material is that the graphite is spherical in the as cast state, its properties are similar to the alloy infinite cold hardening roll, and its strength is higher than the infinite cold hardening roll.

化学成分(Cheical Compositions) wt. %

材质代码(Material code)	C	Si	Mn	Cr	Ni	Mo	Mg
GB/T 1504							
SG II	2.9~3.6	0.8~2.5	0.4~1.2	0.2~0.6	—	0.2~0.6	≥0.04
SG IV	2.9~3.6	0.8~2.5	0.4~1.2	0.2~0.6	0.5~1.0	0.2~0.8	≥0.04
SG V	2.9~3.6	0.8~2.5	0.4~1.2	0.3~1.2	1.0~2.0	0.2~0.8	≥0.04

性能及应用(Properties and Applications)

材质代码(Material code)	辊身硬度 (HSD)	抗拉强度 Rm(N/mm²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
GB/T 1504					
SG II	40-55	≥320	φ 1350	35	小型型钢、棒线材和窄带钢精轧机架；热轧薄板平整轧机 Finishing mill stands for small section steel, bar and wire rod and narrow strip steel; Hot rolled sheet temper mill
SG IV	45-60	≥320			
SG V	50-65	≥320			



NCC轧辊 / Non Continuous Carbides Rolls

代码(Grade): NCC



粗轧机架用轧辊需要一些特性相互结合，其中某些特性会相互抵消对方的作用，这些特性包括耐磨性、耐热裂性、耐冲击性、热硬度和热强度等。网状碳化物能提高耐磨性、耐冲击性。但如果网状碳化物含量过多会严重降低断裂韧性。这在粗轧情况下，会促使热裂纹形成。要减少过多碳化物的影响，又能保持耐磨性，就要加入镍、钼等合金元素，使基体形态为贝氏体/马氏体（针状），比通常的珠光体基体更耐磨。钼元素还有助于提高轧辊高温硬度。

NCC轧辊就是采用高合金材质加上特殊热处理制造出来的，基体组织中碳化物的含量小于5%，满足粗轧机架要求。

Rollers for roughing stands need to combine some characteristics, some of which will counteract each other, including wear resistance, heat crack resistance, impact resistance, thermal hardness and thermal strength. The network carbide can improve the wear resistance and impact resistance. However, if the content of network carbide is too much, the fracture toughness will be seriously reduced. This will promote the formation of hot cracks in rough rolling. In order to reduce the influence of excessive carbides and maintain wear resistance, alloy elements such as nickel and molybdenum should be added to make the matrix form bainite/martensite (acicular), which is more wear-resistant than ordinary pearlite matrix. Molybdenum also helps to improve the high temperature hardness of rolls.

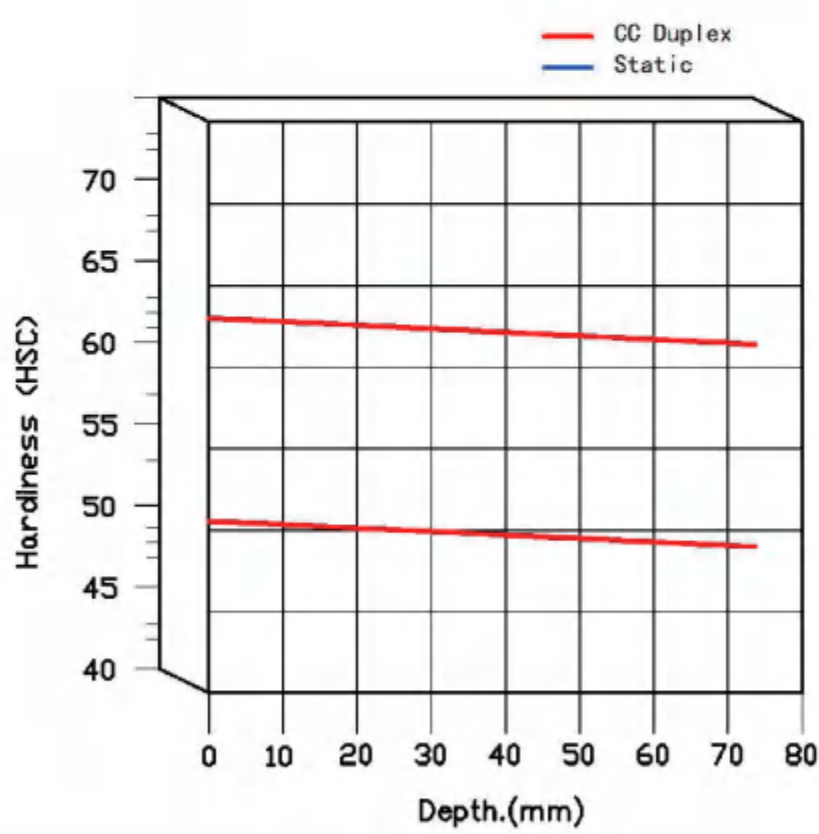
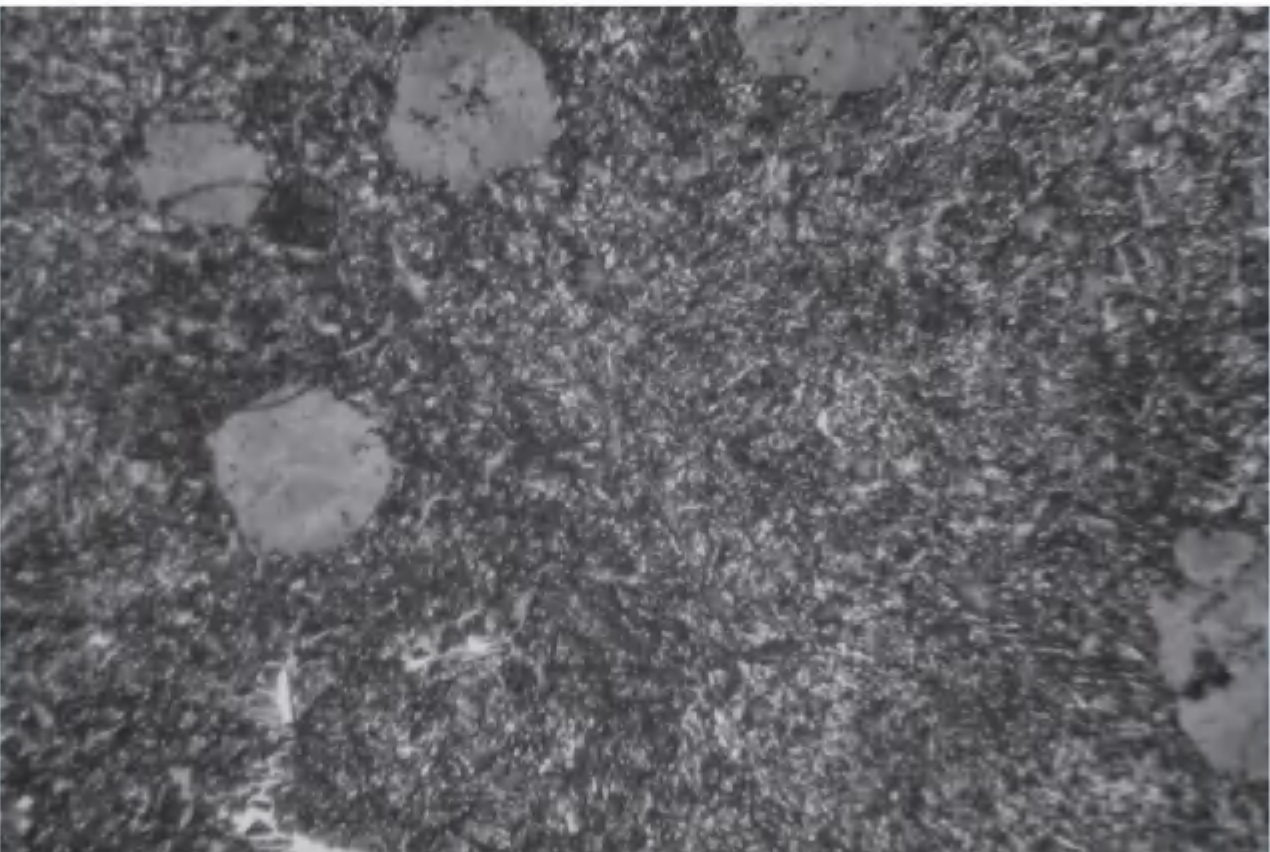
NCC roll is made of high alloy materials and special heat treatment. The content of carbide in the matrix is less than 5%, which meets the requirements of roughing mill stand.

化学成分(Chemical Compositions) wt. %

材质代码(Material code)	C	Si	Mn	Cr	Ni	Mo
自有材质 (Own material)						
NCC I	3.1-3.5	1.5-2.5	0.5-1.0	≤0.15	2.0-3.0	0.4-0.7
NCC II	3.1-3.5	1.5-2.5	0.5-1.0	≤0.15	2.5-4.0	0.5-1.0

性能及应用(Properties and Applications)

材质代码(Material code)	辊身硬度 (HSD)	抗拉强度 Rm(N/mm²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
自有材质 (Own material)					
NCC I	40-50	≥520	φ 1350	35	棒线材粗轧机架、大型型钢BD2辊 Rod and wire roughing stand, large section steel BD2 roll
NCC II	50-60	≥520			



高强度合金球墨铸铁轧辊 / High Strength Alloy Nodular Cast Iron Rolls

代码(Grade): SHD



采用激冷浇注，并通过特殊热处理的合金球墨铸铁轧辊，具有强度高，碳化物分布均匀，硬度落差小，抗热裂性好的优点。适用于冷却条件较差，变形抗力较大或轧制力较大的合金钢粗、中轧机架。

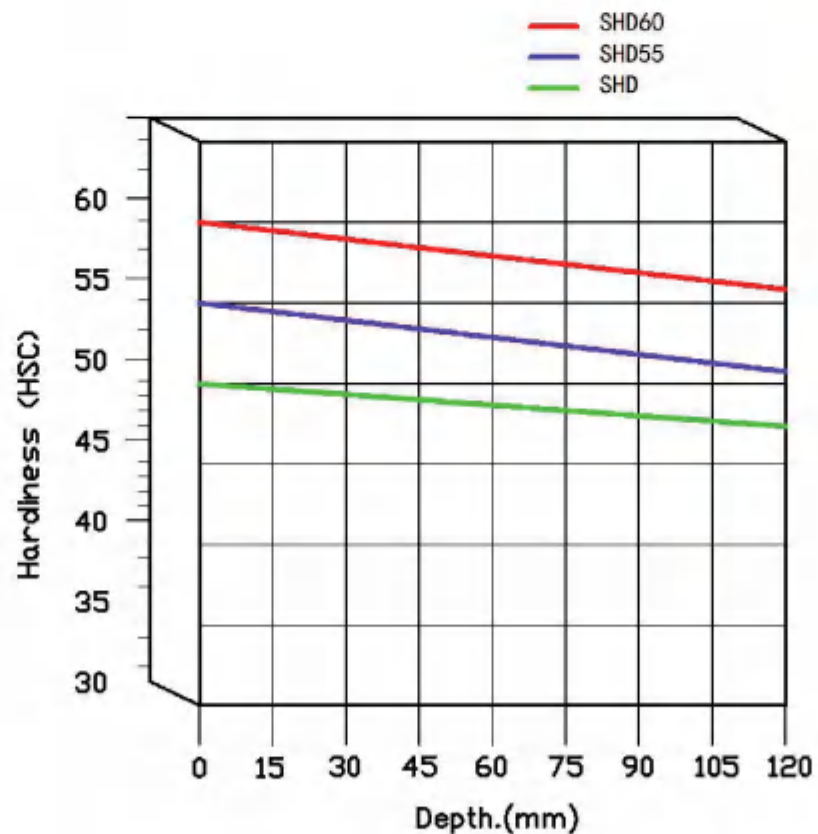
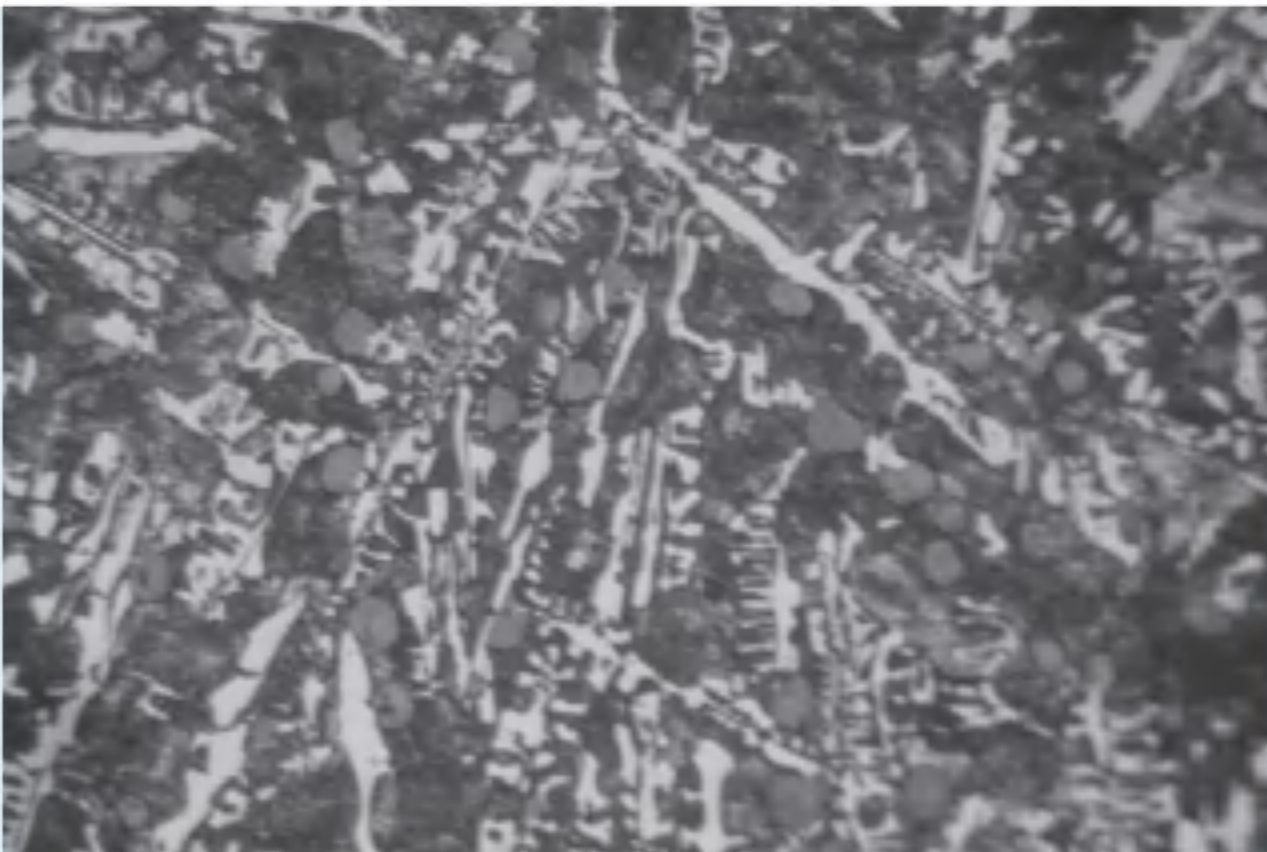
The alloy nodular cast iron roller, which is poured by chilling and has been specially heat treated, has the advantages of high strength, uniform carbide distribution, small hardness drop and good hot crack resistance. It is suitable for coarse and medium rolling stands of alloy steel with poor cooling conditions, large deformation resistance or large rolling force.

化学成分(Chemical Compositions) wt. %

材质代码(Material code)	C	Si	Mn	Cr	Mo	Ni
自有材质 (Own material)						
SHD50	3.1-3.6	1.2-2.0	0.5-1.0	0.1-0.5	0.3-0.7	1.5-2.0
SHD55	3.1-3.6	1.2-2.0	0.5-1.0	0.4-0.8	0.5-1.0	2.0-2.5
SHD60	3.1-3.6	1.2-2.0	0.5-1.0	0.6-1.2	0.5-1.0	2.5-3.0

性能及应用(Properties and Applications)

材质代码(Material code)	辊身硬度 (HSD)	抗拉强度 Rm(N/mm²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
自有材质 (Own material)					
SHD50	40-55	500-600	φ 1350	35	型钢粗、中轧；带钢粗轧； Rough and medium rolling of section steel; Rough rolling of strip steel;
SHD55	50-60	500-600			
SHD60	55-65	450-550			



贝氏体球墨铸铁轧辊 / Bainitic Nodular Cast Iron Rolls

代码(Grade): SGA



贝氏体球铁材质中加入了镍、锰、铬、钼等合金元素，它具有针状组织(贝氏体+少量马氏体)基体，比珠光体球铁轧辊强度更高，韧性更好，耐磨性也明显提高。可采用静态铸造或离心铸造生产。贝氏体球铁轧辊适用于小型型钢连轧、棒材中轧、扁钢精轧、高速线材预精轧、方/板坯初轧。

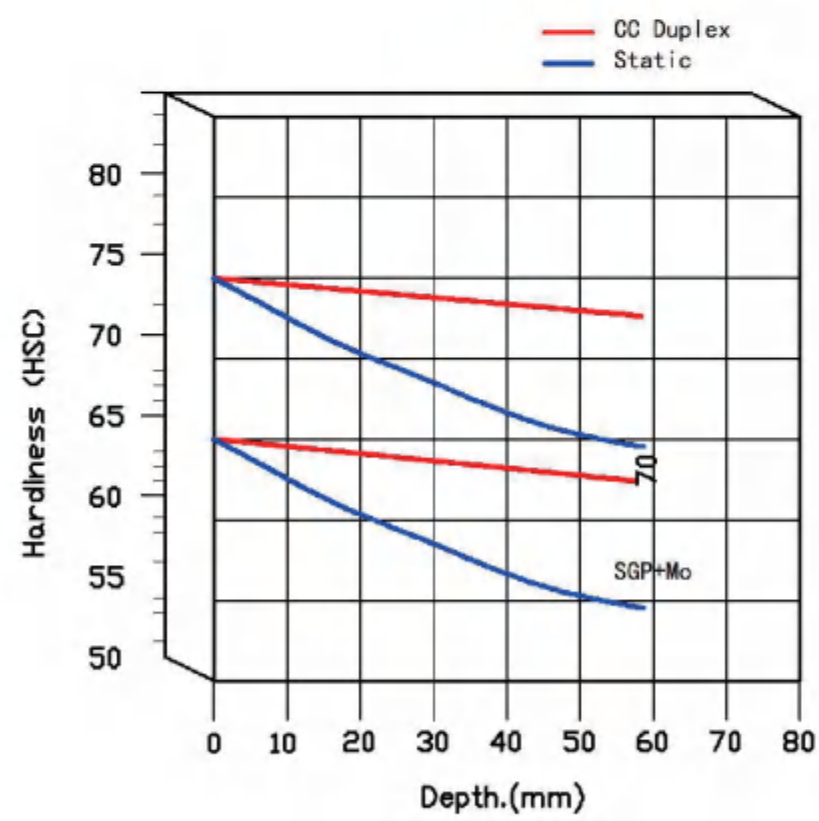
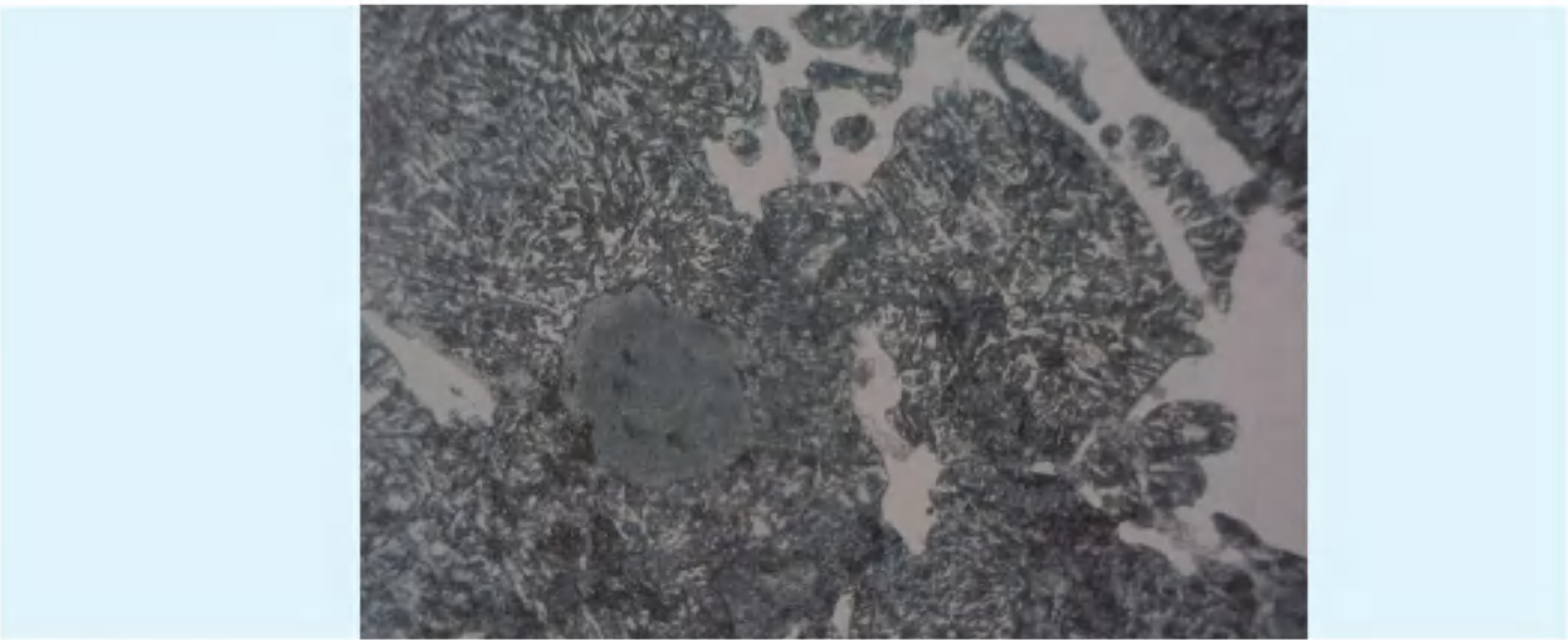
Nickel, manganese, chromium, molybdenum and other alloy elements are added into the bainite ductile iron material. It has an acicular structure (bainite+a small amount of martensite) matrix, which has higher strength, better toughness and significantly improved wear resistance than pearlite ductile iron rolls. It can be produced by static casting or centrifugal casting. The bainite ductile iron roll is suitable for continuous rolling of small section steel, medium rolling of bar, finishing rolling of flat steel, pre finishing rolling of high-speed wire rod, and primary rolling of square/slab.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)	C	Si	Mn	Cr	Ni	Mo
GB/T 1504						
SGA I	2.9-3.6	1.2-2.2	0.2-0.8	0.2-1.0	3.01-3.50	0.5-1.0
SGA II	2.9-3.6	1.0-2.0	0.2-0.8	0.3-1.5	3.51-4.50	0.5-1.0

性能及应用 (Properties and Applications)

材质代码 (Material code)	辊身硬度 (HSD)	抗拉强度 Rm(N/mm²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
GB/T 1504					
SGA I	55-78	≥350	φ 1350	35	小型型钢连轧、棒材中轧、扁钢精轧、高速线材预精轧、方/板坯初轧； Continuous rolling of small section steel, medium rolling of bar, finishing rolling of narrow strip steel High speed wire pre finishing rolling, square / slab blooming;
SGA II	60-80	≥350			



珠光体球墨铸铁轧辊 / Spheroidal Graphite Pearlite Cast Iron Rolls

代码(Grade): SGP



球墨铸铁中加入适量的镍、铬、钼合金元素，经过特殊热处理得到珠光体球铁轧辊。珠光体球墨铸铁轧辊具有良好的强度、高温性能和抗事故性能，工作层硬度落差小。适用于大型初轧机、型钢连轧、棒线材粗、中轧。

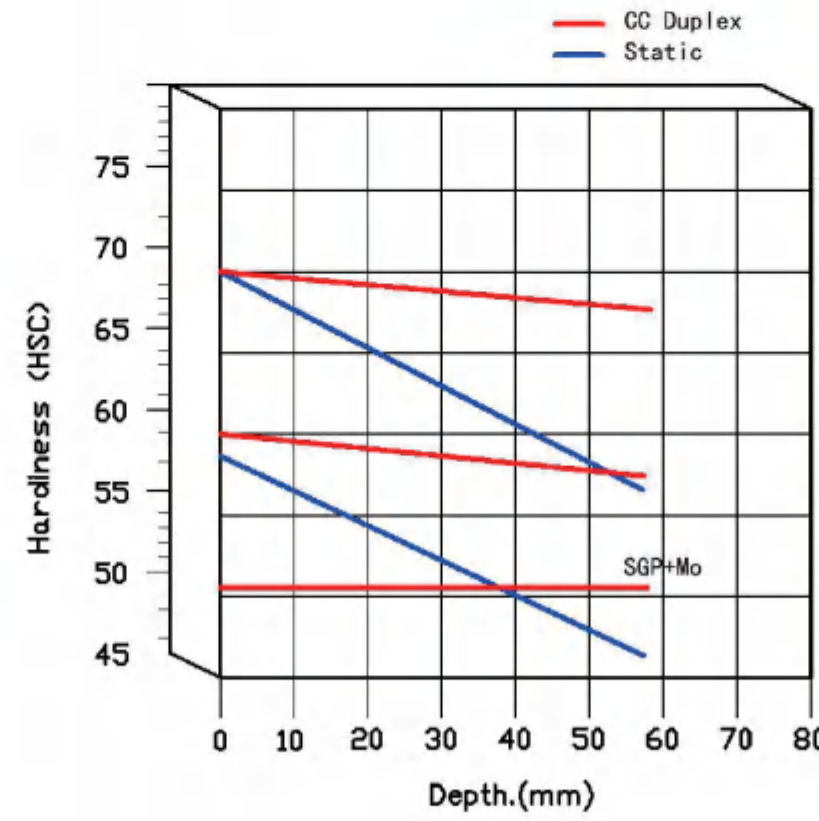
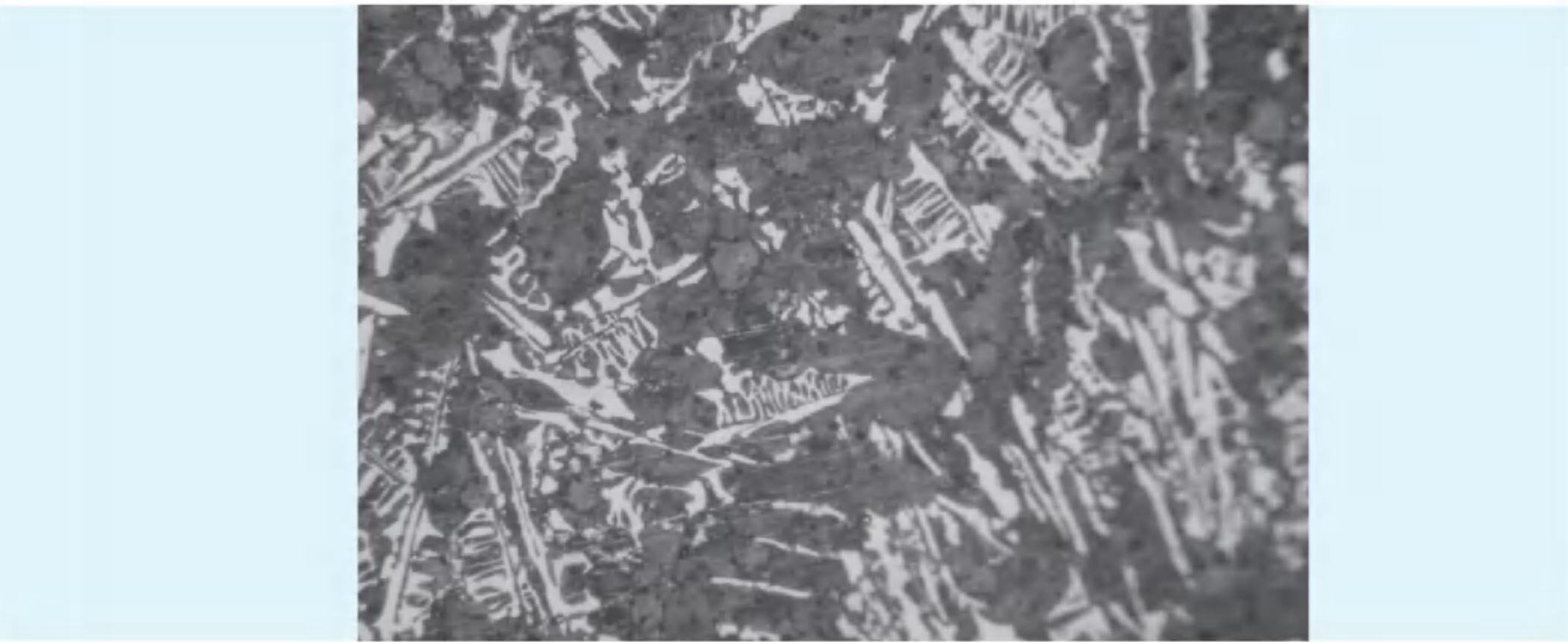
Pearlitic ductile iron roller is obtained by adding proper amount of nickel, chromium and molybdenum alloy elements into nodular cast iron after special heat treatment. Pearlitic ductile iron roll has good strength, high temperature performance and accident resistance, and the hardness drop of working layer is small. Suitable for large blooming mill, continuous rolling of section steel, rough and medium rolling of bar and wire.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)	C	Si	Mn	Cr	Ni	Mo	Mg
GB/T 1504							
SGP I	2.90-3.60	1.40-2.20	0.40-1.00	0.10-0.60	1.50-2.00	0.20-0.80	≥0.04
SGP II	2.90-3.60	1.20-2.00	0.40-1.00	0.20-1.00	2.00-2.50	0.20-0.80	≥0.04
SGP III	2.90-3.60	1.00-2.00	0.40-1.00	0.20-1.20	2.50-3.00	0.20-0.80	≥0.04

性能及应用 (Properties and Applications)

材质代码 (Material code)	辊身硬度 (HSD)	抗拉强度 Rm(N/mm²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
GB/T 1504					
SGP I	45-60	≥450	φ 1350	35	大型初轧机、型钢连轧、棒线材粗、中轧 Large blooming mill, continuous rolling of section steel, coarse and medium rolling of rod and wire rod
SGP II	50-65	≥450			
SGP III	55-75	≥450			



高铬铸铁轧辊 / High Chromium Cast Iron Rolls

代码(Grade): HCr



离心复合高铬铸铁轧辊是高合金轧辊，其工作层基体组织为马氏体+屈氏体/索氏体，碳化物主要是板条状和菊花状的Cr₇C₃型共晶碳化物。芯部采用高强度球墨铸铁材质。也可以采用离心或常法铸造生产高铬铸铁辊环。适用于热轧带钢轧机精轧辊，棒材中、精轧辊，万能轧机辊环。

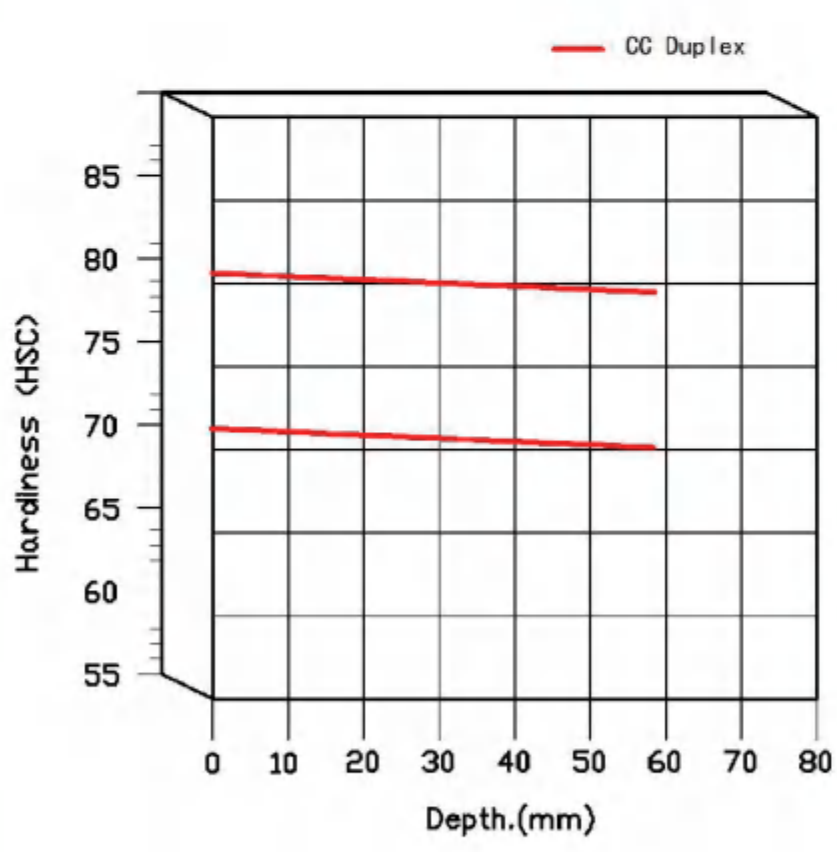
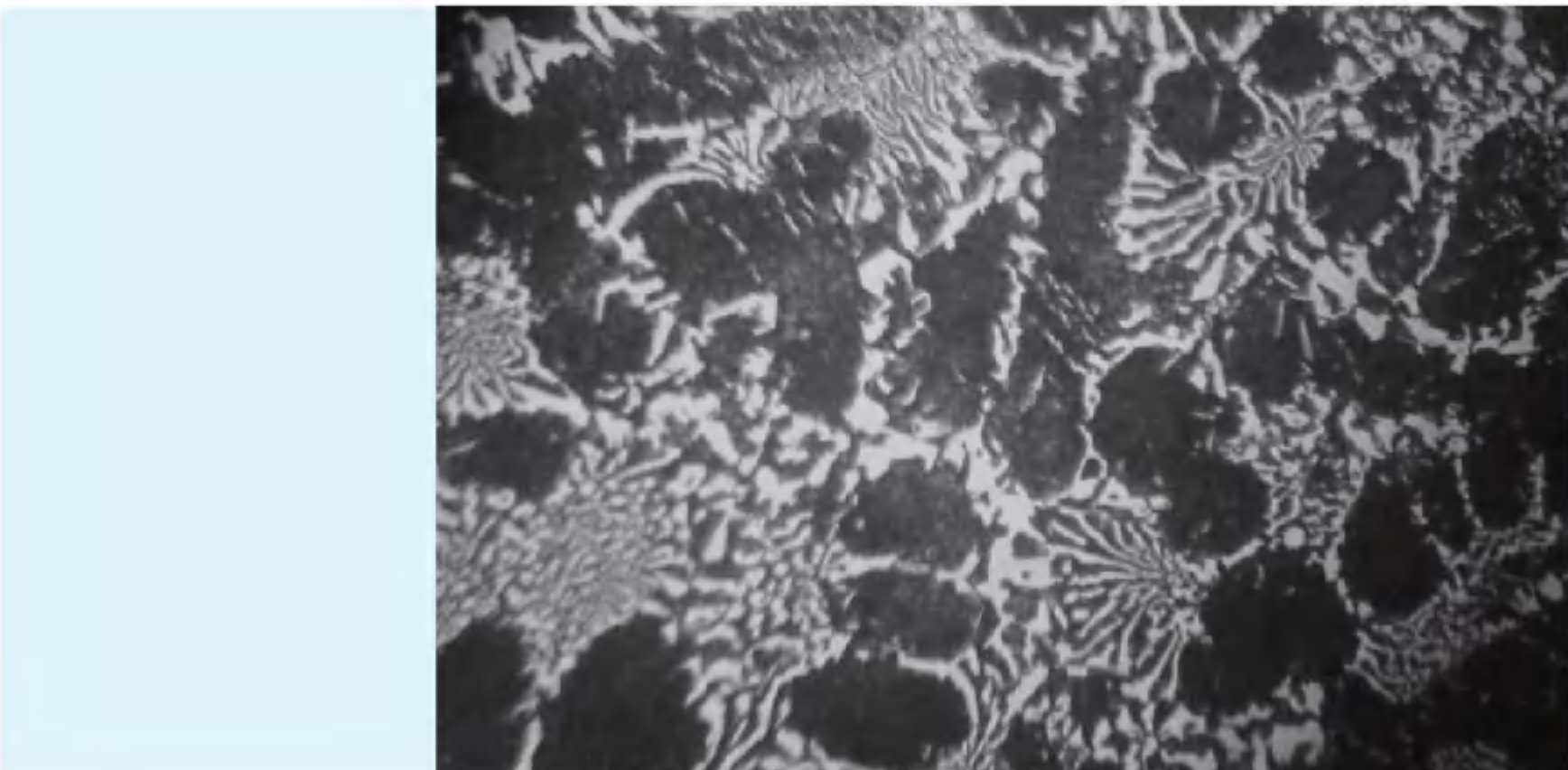
The centrifugal composite high chromium cast iron roll is a high alloy roll. The matrix structure of its working layer is martensite+troostite/sorbite, and the carbides are mainly plate like and chrysanthemum like Cr₇C₃ eutectic carbides. The core is made of high-strength ductile iron. High chromium cast iron roll rings can also be produced by centrifugal casting or normal casting. It is suitable for finishing roll of hot strip mill, intermediate and finishing roll of bar, and roll ring of universal mill.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)	C	Si	Mn	Cr	Ni	Mo	V
GB/T 1504							
HCr I	2.3-3.3	0.3-1.0	0.5-1.2	12.00-15.00	0.7-1.7	0.70-1.50	0.0-0.6
HCr II	2.3-3.3	0.3-1.0	0.5-1.2	15.01-18.00	0.7-1.7	0.70-1.50	0.0-0.6
HCr III	2.3-3.3	0.3-1.0	0.5-1.2	18.01-22.00	0.7-1.7	1.51-3.00	0.0-0.6

性能及应用 (Properties and Applications)

材质代码 (Material code)	辊身硬度 (HSD)	抗拉强度 R _m (N/mm ²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
GB/T 1504					
HCr I	60-75	≥350	φ 1700	20	热轧带钢轧机精轧辊；棒材中、精轧辊； 万能轧机辊环； Finishing roll of hot strip mill; Bar intermediate and finishing rolls; Universal rolling mill roll ring;
HCr II	65-80	≥350			
HCr III	75-90	≥350			



合金无限冷硬铸铁轧辊 / Alloy Indefinite Chilled Cast Iron Rolls

代码(Grade): IC



合金无限冷硬轧辊工作层中有细小晶间石墨。石墨和碳化物的大小、形状、分布可通过激冷作用和合金含量来控制，由于添加了锰、铬，钼等合金元素，基体组织可以是珠光体、贝氏体或马氏体。加上有少量细小石墨存在，不仅提高了轧辊抗剥落性、抗热裂性和耐磨性能，而且辊身工作层硬度落差很小。使用于棒、线材、小型型钢轧机中轧、精轧机架。

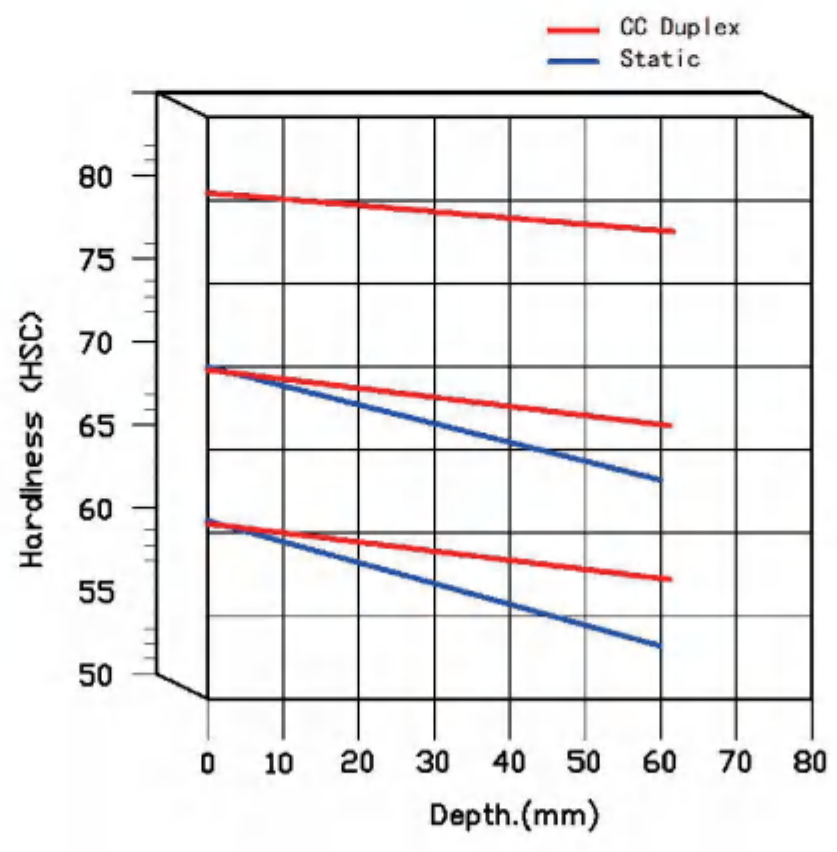
There is fine intergranular graphite in the working layer of alloy infinitely chilled roll. The size, shape and distribution of graphite and carbide can be controlled by chilling effect and alloy content. As manganese, chromium, nickel, molybdenum and other alloy elements are added, the matrix structure can be pearlite, bainite or martensite. In addition, there is a small amount of fine graphite, which not only improves the anti stripping, anti hot cracking and wear resistance of the roll, but also reduces the hardness drop of the working layer of the roll body. It is used for medium and finish rolling stands of bar, wire rod and small section steel mills.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)	C	Si	Mn	Cr	Ni	Mo	V	W	Nb
GB/T 1504									
IC II	2.9-3.6	0.6-1.2	0.4-1.2	0.7-1.2	1.01-2.00	0.2-0.6	—	—	—
IC III	2.9-3.6	0.6-1.2	0.4-1.2	0.7-1.2	2.01-3.00	0.2-1.0	—	—	—
IC IV	2.9-3.6	0.6-1.5	0.4-1.2	1.0-2.0	3.01-4.80	0.2-1.0	—	—	—
IC V	2.9-3.6	0.6-1.5	0.4-1.2	1.0-2.0	3.01-4.80	0.2-2.0	0.2-2.0	0.0-2.0	0.0-2.0

性能及应用 (Properties and Applications)

材质代码 (Material code)	辊身硬度 (HSD)	抗拉强度 R _m (N/mm ²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
GB/T 1504					
IC II	55-70	≥160	φ 800	15	中小型钢和窄带钢中、精轧机架 Medium and finish rolling stands for small and medium section steel and narrow strip steel
IC III	60-75	≥350			
IC IV	65-80	≥350			中厚板、热带钢轧机工作辊、平整辊 Working roll and leveling roll of plate and hot strip mill
IC V	70-85	≥350			



高铬铸钢轧辊 / High Chromium Cast Steel Rolls

代码(Grade): HCrS



高铬铸钢轧辊硬度高，其基体组织为回火马氏体+碳化物+少量残余奥氏体。碳化物主要为M7C3、M2C型，耐磨性和抗热裂性能好。碳化物含量较低，一般10%左右，对轧机无特殊水冷却要求，轧制中自身形成氧化膜，有利于提高轧材表面质量。高铬钢轧辊主要适用于热轧带钢粗轧辊，也可用于型钢轧机中、精轧轧辊、水平辊环、立辊环。

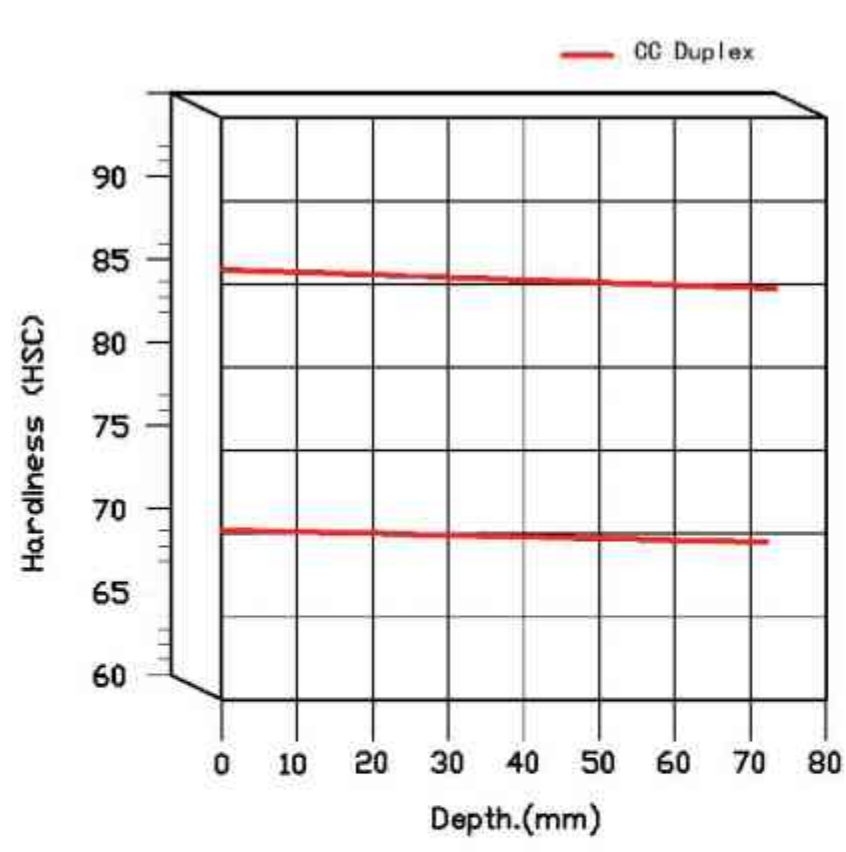
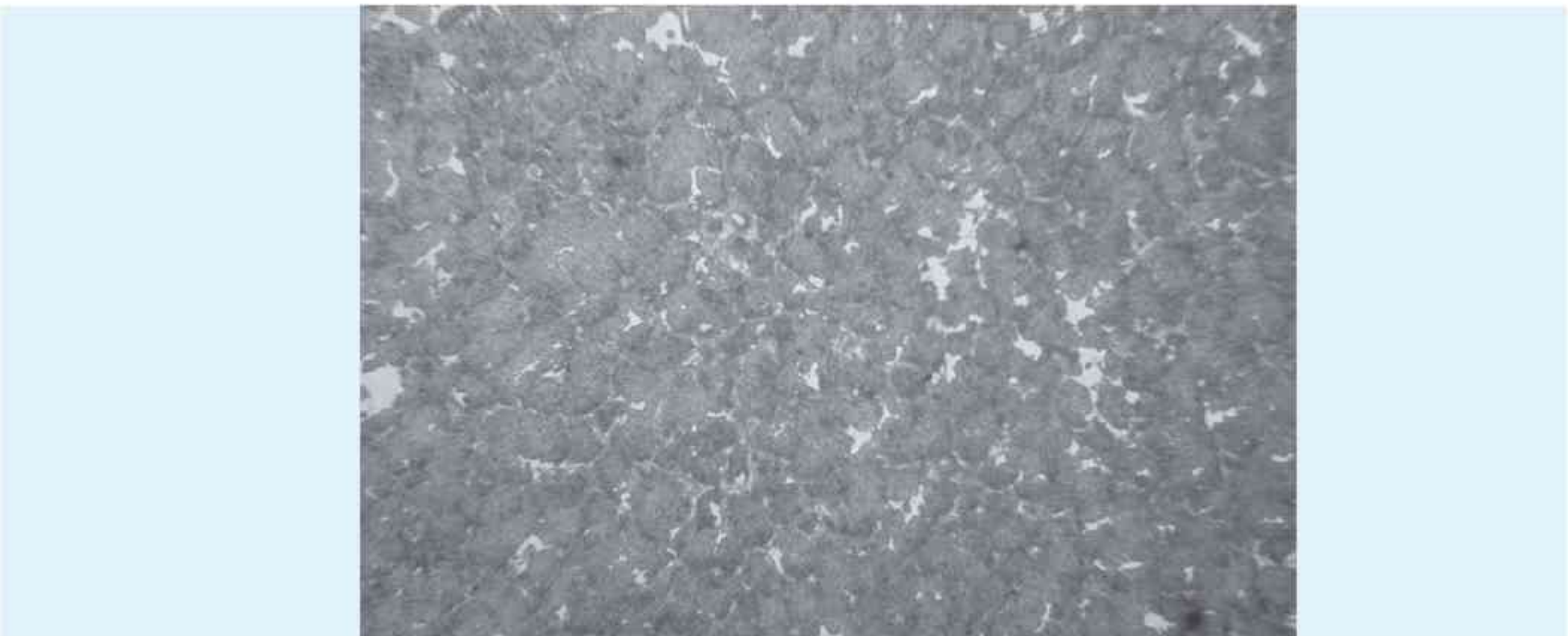
The high chromium cast steel roll has high hardness, and its matrix structure is tempered martensite+carbide+a small amount of retained austenite. The carbides are mainly M7C3 and M2C, with good wear resistance and hot cracking resistance. The carbide content is low, generally about 10%. There is no special water cooling requirement for the rolling mill. The oxide film formed during rolling is conducive to improving the surface quality of rolled products. The high chromium steel roll is mainly used for the rough roll of hot strip steel, as well as the middle and finish roll, horizontal roll ring and vertical roll ring of section steel mill.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)	C	Si	Mn	Cr	Ni	Mo	P	S
YB/T 4906								
HCrS-350	1.2-2.0	0.3-1.0	0.5-1.5	8.0-14.0	0.0-1.5	0.8-2.0	≤0.03	≤0.03
HCrS-600	0.6-1.6	0.3-1.0	0.5-1.5	3.0-8.0	0.0-1.5	0.8-2.0	≤0.03	≤0.03

性能及应用 (Properties and Applications)

材质代码 (Material code)	辊身硬度 (HSD)	抗拉强度 Rm(N/mm²)	最大规格 (mm)	最大单重 (t)	推荐用途 (Recommended use)
YB/T 4906					
HCrS-350	65-75	≥350	φ 800	15	热轧带钢粗轧辊、立辊； 型钢轧机用中、精轧辊、水平辊环、立辊环； Rough roll and vertical roll of hot rolled strip; Intermediate and finishing rolls, horizontal roll rings and vertical roll rings for section mills;
HCrS-600	55-65 70-85	≥600 ≥600			



锻钢轧辊、辊轴 / Forged Steel Rolls, Roll Shaft



合金锻钢轧辊具有优异的综合轧制性能，如高耐热疲劳性能、高耐冲击性能和良好的抗裂纹扩展性能，辊颈部位具有高韧性和高抗拉强度。锻钢轧辊广泛用于大、中型型钢轧机粗轧辊、矫直辊。

锻钢辊轴与各种材质辊环装配组合成各种型钢轧机用辊。

Alloy forged steel rolls have excellent comprehensive rolling properties, such as high heat fatigue resistance, high impact resistance and good crack growth resistance. The roll neck has high toughness and high tensile strength. Forged steel roll is widely used for rough roll and straightening roll of large and medium-sized section mill.

Forged steel roll shaft and roll rings of various materials are assembled to form rolls for various section steel mills.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)	C	Si	Mn	Cr	Ni	Mo	P	S
YB/T 4906								
9Cr2Mo	0.85-0.95	0.25-0.45	0.2-0.35	1.7-2.1	≤0.25	0.2-0.4	≤0.03	≤0.03
FS-800 I	0.4-0.7	0.15-0.45	0.6-1.0	0.6-1.0	0.0-0.5	0.15-0.40	≤0.035	≤0.025
FS-800 II	0.4-0.7	0.15-0.45	0.6-1.0	0.6-1.0	0.5-1.5	0.2-0.6	≤0.035	≤0.025
42CrMo	0.38-0.45	0.17-0.37	0.5-0.8	0.9-1.2	—	0.15-0.30	≤0.03	≤0.03
60CrMnMo	0.55-0.65	0.20-0.45	0.9-1.2	0.8-1.2	—	0.20-0.45	≤0.03	≤0.03
60CrNiMo	0.55-0.65	0.2-0.6	0.7-1.2	0.8-1.2	0.2-1.5	0.2-0.6	≤0.03	≤0.03
36NiCrMo16	0.32-0.39	≤0.4	0.3-0.6	1.6-2.0	3.6-4.1	0.25-0.45	≤0.03	≤0.03

性能及应用 (Properties and Applications)

材质代码 (Material code)	辊身硬度 (HSD)	抗拉强度 Rm(N/mm²)	推荐用途 (Recommended use)
YB/T 4906			
9Cr2Mo	55-70	≥600	型钢轧机用粗轧辊、矫直辊 Rough roll and straightening roll for section mill
FS-800 I	35-45	≥800	
FS-800 II	35-45	≥800	大中型型钢轧机用粗轧辊 Coarse roll for large and medium section mill
42CrMo	37-47	≥800	
60CrMnMo	37-47	≥800	型钢轧机用辊轴 Roll shaft for section mill
60CrNiMo	37-47	≥800	
36NiCrMo16	45-55	≥950	

轧边辊 / Edger Rolls



轧边辊是万能机组中，保证宽度及翼缘表面质量的主要备件，要求轧边辊有较高的硬度和较小的硬度落差，不同规格的型钢轧边辊可以选择不同材质和硬度。

The edging roll is the main spare part of the universal unit to ensure the width and flange surface quality. The edging roll is required to have a higher hardness and a smaller hardness drop. Different materials and hardness can be selected for different specifications of section steel edging rolls.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)	C	Si	Mn	Cr	Ni	Mo	V
SGP	3.1-3.5	1.0-2.0	0.5-1.0	0.3-1.3	2.0-3.0	0.4-1.0	—
SGA	3.1-3.5	1.0-2.0	0.2-0.8	0.3-1.3	3.0-4.0	0.6-1.2	—
AD	1.7-2.3	0.3-1.0	0.5-1.5	1.0-3.0	1.0-2.0	0.4-1.0	—
HCr	2.5-3.0	0.5-1.0	0.5-1.5	12.0-18.0	0.5-1.5	1.0-2.0	—
GS	1.6-2.5	1.2-2.0	0.5-1.6	1.0-3.0	1.0-3.0	0.4-1.5	—
HWD	1.4-1.6	0.1-0.6	0.3-0.8	10.0-14.0	—	0.7-1.2	0.8-1.5

性能及应用 (Properties and Applications)

材质代码 (Material code)	辊身硬度 (HSD)	抗拉强度 R_m (N/mm ²)	推荐用途 (Recommended use)
SGP	60-70	400	H型钢, T型钢, 铁轨, 槽钢、工字钢等 H-section steel, T-section steel, rail, channel steel, I-beam, etc
SGA	65-75	400	
AD	50-60	450	
HCr	60-70	600	
GS	50-70	500	
HWD	55-65	500	

万能轧机粗、精轧用辊环 / Roll ring for rough and finish rolling of universal mill



根据使用要求不同选择各种材质生产辊环，如珠光体、贝氏体、半钢、GT半钢、石墨钢、高铬铁、高铬钢、高速钢等。

将辊环通过过盈配合装配到高强度锻钢辊轴上，在保证工作层耐磨的同时有足够的强度，同时辊轴可重复使用多次，节约能源和成本。

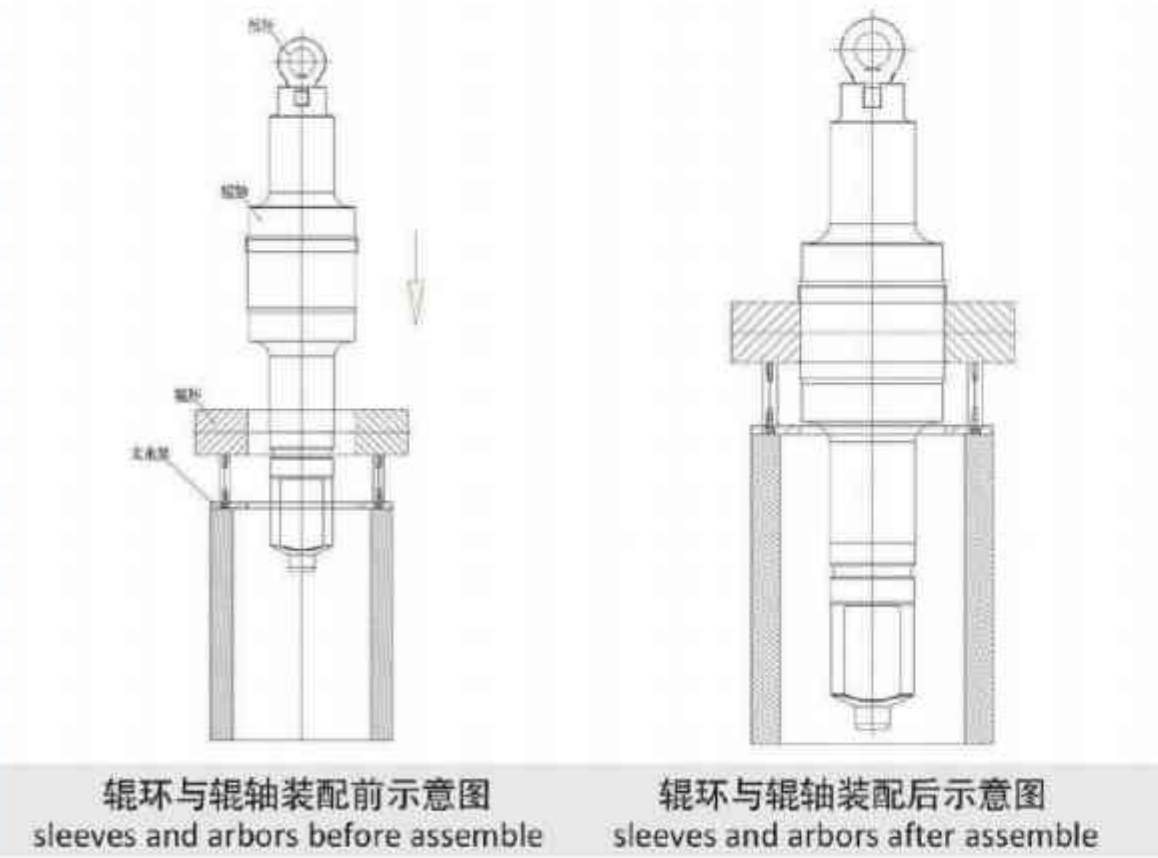
Various materials are selected to produce roll rings according to different requirements, such as pearlite, bainite, semi steel, GT semi steel, graphite steel, high chromium iron, high chromium steel, high-speed steel, etc. The roll ring is assembled on the high-strength forged steel roll shaft through interference fit, which ensures the wear-resistance of the working layer and has sufficient strength. At the same time, the roll shaft can be reused for many times, saving energy and cost.

化学成分 (Chemical Compositions) wt. %

材质代码 (Material code)	C	Si	Mn	Cr	Ni	Mo	V	W	Nb	B
SGP	3.0-3.5	1.0-2.2	0.4-1.0	0.2-1.2	1.5-3.0	0.2-0.8	—	—	—	—
SGA	3.0-3.5	1.0-2.2	0.2-0.8	0.2-1.5	3.0-4.5	0.5-1.0	—	—	—	—
SHD	3.1-3.6	1.2-2.0	0.5-1.5	0.2-1.0	1.5-3.0	0.3-1.2	—	—	—	—
HCr	2.5-3.0	0.5-1.0	0.5-1.5	12-18	0.5-1.5	1.0-2.0	—	—	—	—
AD	1.5-2.1	0.3-0.8	0.6-1.3	0.8-3.5	0.5-2.5	0.2-0.8	—	—	—	—
GT	1.4-1.6	0.3-0.8	0.7-1.5	1.0-2.0	1.0-3.0	0.4-0.8	—	—	—	—
GS	1.6-2.5	1.2-2.0	0.5-1.6	1.0-3.0	1.0-3.0	0.4-1.5	—	—	—	—
IC	2.9-3.6	0.6-1.2	0.4-1.2	0.7-2.0	1.0-4.8	0.2-1.0	—	—	—	—
HCrS	0.6-1.6	0.3-1.0	0.5-1.5	3.0-8.0	0.0-1.5	0.8-2.0	—	—	—	—
HSS	1.5-2.5	0.3-1.5	0.3-1.0	2.0-8.0	0.0-1.5	3.0-8.0	2.0-8.0	0.0-4.0	—	—
S-HSS	0.5-1.5	0.8-1.5	0.5-1.0	3.0-9.0	0.2-1.2	2.0-5.0	0.4-3.0	0.0-3.0	0.0-2.0	—
B-HSS	2.0-3.0	0.3-1.5	0.4-1.2	10.0-17.0	1.0-3.0	0.5-2.0	0.1-1.0	—	—	0.1-1.0

性能及应用 (Properties and Applications)

材质代码 (Material code)	辊身硬度 (HSD)	抗拉强度 R_m (N/mm ²)	推荐用途 (Recommended use)
SGP	55-65	450	型钢万能轧机水平辊环、立轧辊环、棒线材 粗中轧辊环、矫直辊环 Horizontal roll ring, vertical roll ring Coarse and medium roll ring and straightening roll ring of rod and wire
SGA	60-70	450	
SHD	55-65	500	
HCr	60-70	600	
AD	50-60	600	
GT	55-65	700	
GS	50-70	500	
IC	55-85	350	
HCrS	70-85	600	
HSS	70-90	400	
S-HSS	70-85	400	
B-HSS	75-85	400	



辊环与辊轴装配前示意图
sleeves and arbors before assemble

辊环与辊轴装配后示意图
sleeves and arbors after assemble

矫直辊 / Straightening Rolls



矫直辊用于型钢等各种材料在轧制过程中或轧制结束后的矫直工作。矫直机是轧制生产线上的重要设备，保证了生产出高平整度、平直度、尺寸精确的产品。

The straightening roll is used for straightening various materials such as section steel during or after rolling. Straightener is an important equipment on the rolling production line, which ensures the production of products with high flatness, straightness and accurate size.

化学成分(Chemical Compositions) wt. %

材质名称 (Material name)	材质代码 (Material code)	C	Si	Mn	Cr	Ni	Mo	P	S	Mg
合金锻钢	9Cr2Mo	0.85~0.95	0.25~0.45	0.2~0.35	1.7~2.1	≤0.25	0.2~0.4	≤0.03	≤0.03	—
合金半钢	AD	1.5~2.1	0.3~0.8	0.6~1.0	0.7~1.7	0.5~1.5	0.2~0.6	≤0.03	≤0.03	—
石墨钢	GS	1.6~2.1	1.0~1.8	0.6~1.6	1.0~2.0	1.0~2.0	0.4~1.0	≤0.03	≤0.03	—
合金球铁	DC	3.1~3.6	1.2~2.0	0.3~1.0	0.3~1.3	2.5~4.5	0.4~1.4	≤0.06	≤0.03	≥0.04
无限冷硬铸铁	IC	3.1~3.6	0.5~1.5	0.3~1.0	0.8~1.8	2.5~4.5	0.3~0.8	≤0.06	≤0.03	—

性能及应用(Properties and Applications)

材质代码(Material code)	表面硬度(HSD)	抗拉强度Rm(N/mm²)	推荐用途(Recommended use)
9Cr2Mo	70~90	≥600	型钢轧机用矫直辊 Straightening roll for section mill
AD	40~50; 50~60	≥500	
GS	40~50; 50~60	≥500	
DC	55~70	≥350	
IC	65~75	≥200	

凯达产品技术参数 / Technical parameters of Kaida products

特色产品简介 / Introduction To Featured Products



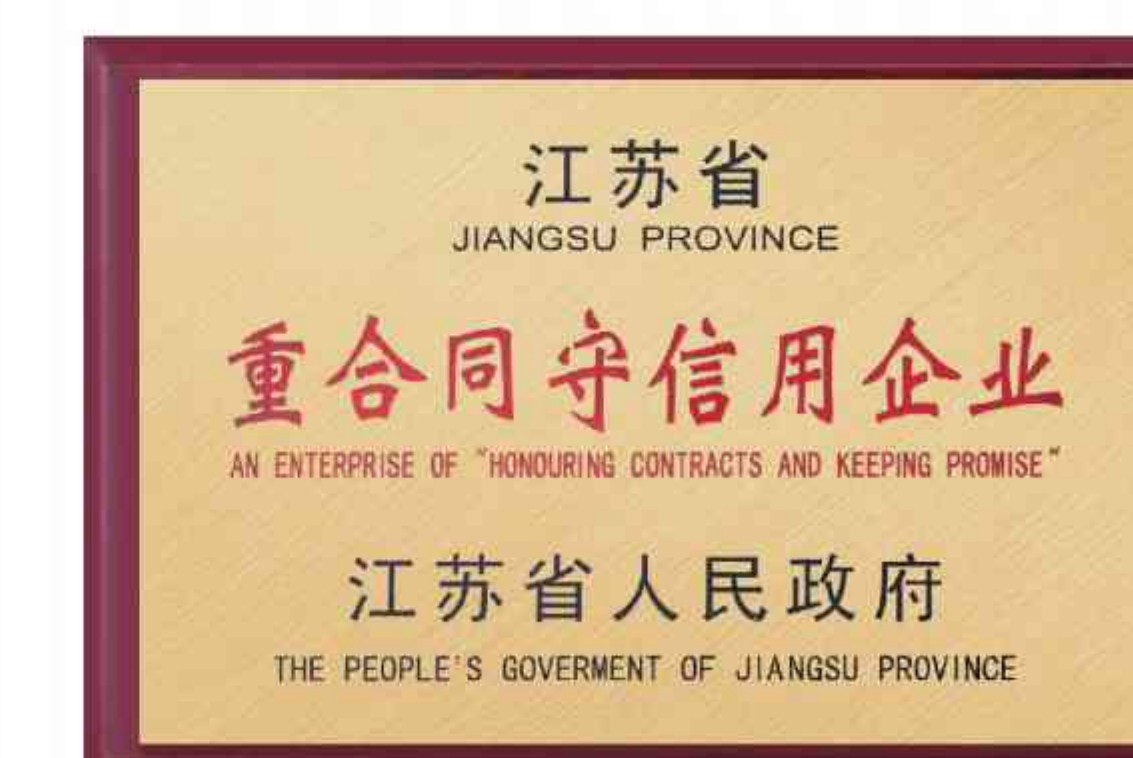
凯达通过40多年轧辊生产经验和业绩，在广大用户的支持帮助下，通过技术引进、自主研发形成了部分凯达质量稳定的特色产品，在使用过程中得到客户良好的评价。

Through more than 40 years of roll production experience and performance, with the support and help of users, Kaida has formed some characteristic products with stable quality through technology introduction and independent research and development, which have been well evaluated by customers in the process of use.

分类 (Classification)	材质 (Material Quality)	碳含量% (Carbon content%)	辊身硬度 (HSD)	抗拉强度 Rm(N/mm²)	使用推荐 (Usage Recommendations)
铸钢类轧辊 Cast steel roll	AS	0.5~0.9	38~48	650	大型型钢、棒材粗轧 Rough rolling of large section steel and bar
		0.5~0.9	55~65	800	带钢支承辊、大型钢精轧 Strip steel backup roll, large steel finish rolling
	AD120	1.0~1.2	40~48	650~750	中型型钢、棒材BD Medium section steel, bar BD
	GS200	1.9~2.2	50~65	450~550	型钢精轧、棒材精轧 Section steel finish rolling, bar finish rolling
	GS230	2.2~2.4	65~70	400~550	型钢精轧、万能轧机辊环、大型棒材中精轧 Section steel finish rolling, universal mill roll ring, large bar medium finish rolling
铸铁类轧辊 Cast iron roll	NCC	3.1~3.5	40~55	550~650	大型型钢BD2、棒材、型钢、带钢粗轧 Rough rolling of large section steel BD2, bar, section steel and strip steel
	SHD	3.1~3.5	45~65	500~600	中小型型钢粗中轧、棒材粗中轧 Medium and small section steel rough and medium rolling, bar rough and medium rolling
	HCr	2.5~3.0	60~75	600	型钢水平辊环、立辊环、轧边、棒材精轧 Section steel horizontal roll ring, vertical roll ring, edging, bar finishing rolling

STRENGTH CASTS BRILLIANCE ENTERPRISE HONOR

实力铸就辉煌
企业荣誉





KAIDA'S ACHIEVEMENTS AND VALUES PATENT CERTIFICATE

凯达的成就与价值
相关专利证书



熔炼精品 · 牢记担当责任使命
铸造诚信 · 携手共创美好未来

WE SHOULD BEAR IN MIND OUR RESPONSIBILITY AND MISSION
CASTING INTEGRITY AND WORKING TOGETHER TO CREATE A BETTER FUTURE

凯达重工

